CMOS: References and Regulators

CASCODED BETA-MULTIPLIER BANDGAP REFERENCE

Mini-Project 1 • TSMC 180 nm, 3.3 V devices

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1 Introduction and Specifications

A bandgap reference (BGR) produces a voltage that is nearly independent of supply, process and temperature by adding the emitter–base voltage of a bipolar transistor, which falls with temperature (CTAT), to a voltage proportional to absolute temperature (PTAT) [1]. This report describes a cascoded beta-multiplier BGR in TSMC 180 nm with 3.3 V devices, with a native-NMOS one-transistor start-up and an enable input EN. It was simulated in Cadence Virtuoso (ADE Explorer, ADE Assembler, Spectre; on 10.2.20.27 server, IIIT2024702001 user) at the typical point, over 81 PVT corners and with a 200-point Monte Carlo analysis.

1.1 Specifications

Table 1: Specifications of the bandgap reference.

Parameter	Specification
Technology	TSMC 180 nm, 3.3 V devices
Topology	Cascoded beta-multiplier bandgap reference
Start-up	Native (NAT) one-MOS start-up, pull-up and pull-down transistors, enable signal
Power consumption	$< 10\,\mu W$
Output variation (PVT + mismatch)	$< \pm 8\,\%$
Supply range	$3.3\,V \pm 10\,\%$ (2.97–3.63 V)
Temperature range	$-40\,^\circ C$ to $125\,^\circ C$
PSRR magnitude	$> 40\,dB$

1.2 Verification conditions

PVT: process ss/tt/ff (MOS, native and bipolar corners tied), resistor ss/tt/ff_res, VDD 2.97/3.3/3.63 V and temperature $-40/27/125\,^\circ C$, i.e. 81 corners (27 when temperature or supply is swept in the analysis). **Monte Carlo:** 200 samples, process and mismatch, at tt, 27 °C, 3.3 V, plus four mismatch-only runs. **Accuracy:** PVT minimum -3σ and PVT maximum $+3\sigma$, relative to the typical V_{REF}.

1.3 Assumptions

Where the problem statement leaves a condition open, the assumptions of Table 2 are used.

Table 2: Design and verification assumptions.

Item	Assumption
Power	$< 10\,\mu W$ must hold at every PVT corner, including the highest-current one (FF, ff_res, 3.63 V, 125 °C); hence about 0.5 µA per branch at tt, 27 °C.
Power-down	EN = 0 turns the reference off; target supply current $< 10\,nA$.
Enable level	EN = 1 is a logic high equal to VDD, from the same supply.
PSRR	The $> 40\,dB$ requirement applies at low frequency and is checked at 1 Hz; the frequency where the PSRR falls to 40 dB is also reported.
Output load	No DC load current. 1 fF on V_{REF} in the DC, temperature, PSRR, sequencing and no-start-up testbenches; 1 pF in the start-up testbench, so start-up times are quoted with 1 pF.
Temperature coefficient	End-point method: $TC = [V_{REF}(125\,°C) - V_{REF}(-40\,°C)]/[V_{REF}(27\,°C) \cdot 165\,°C]$.
Minimum supply	From the supply sweep: VDD at which V_{REF} is 1 % below its 3.3 V value. Per branch: the VDD at which the branch cascode leaves saturation, evaluated at VDD = 3.3 V (Section 4.4).
Settling time	Time from the start of the VDD ramp (1 µs; ramp 0 to VDD in 10 µs) until V_{REF} reaches 99 % of its value at 1 ms.
Model artefact	The MOS-model glitch between −36 and −31 °C is ignored, as instructed; it is noted where it affects a value.

1.4 Summary of results

Table 3: Specification versus achieved performance. Typical: tt, 27 °C, 3.3 V. Worst case: over all PVT corners or Monte Carlo samples.

Parameter	Spec	Typical	Worst case (PVT / MC)	Status
Output voltage V_{REF}	–	1.255 V	1.249–1.263 V (PVT); $\sigma = 17.56\,mV$ (MC)	–
Variation	$< \pm 8\,\%$	–	−4.68 % / +4.84 %	Met
Power	$< 10\,\mu W$	5.05 µW	8.977 µW (FF, ff_res, 3.63 V, 125 °C)	Met
Supply range	2.97–3.63 V	LS 678.6 µV/V	LS $\leq 2.505\,mV/V$; $V_{DD,min} \leq 2.207$ V	Met
Temperature range	−40...125 °C	−0.068 ppm/°C	TC −6.0...25.9 ppm/°C	Met
PSRR at 1 Hz	$> 40\,dB$	69.9 dB	$\geq 47.4\,dB$	Met
Start-up	must start	16.11 µs	starts at all 81 corners and 200 samples; $\leq 44.17\,\mu s$	Met
Power-down current	–	42 pA	–	–

2 Circuit Description

2.1 Topology

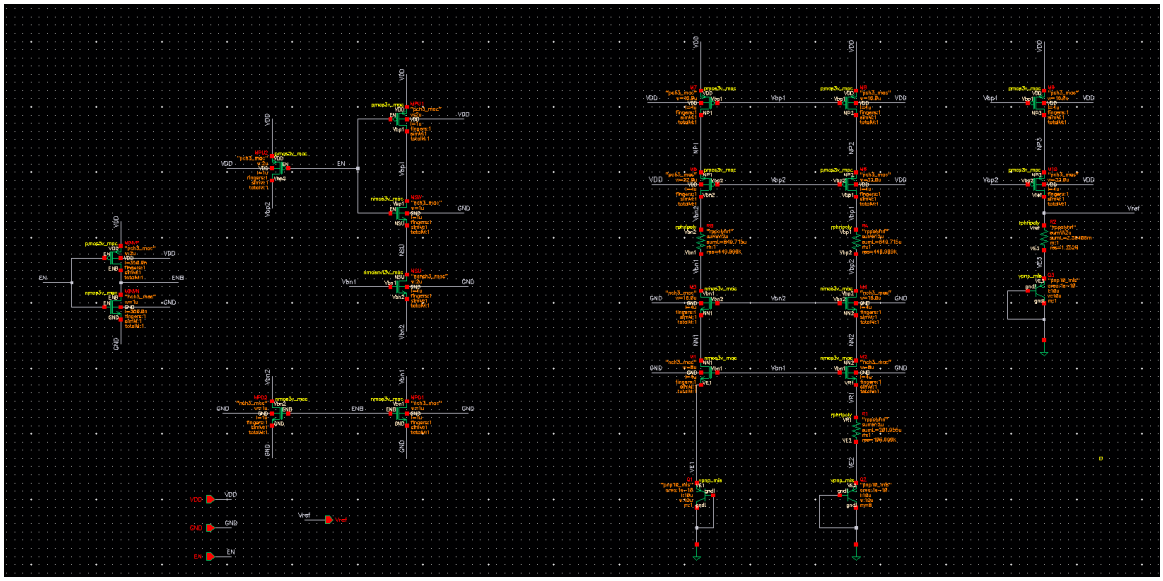


Figure 1: Schematic of the bandgap reference (library `JaiSrikar_CMOS`, cell `mid-project-bgr`): cascoded beta-multiplier core (M1–M8, RA, RB, R1, Q1, Q2), output branch (M9, M10, R2, Q3), native start-up device MSU with switch MSW, pull-ups MPU1/MPU2, pull-downs MPD1/MPD2, EN → ENB inverter and the pins VDD, GND, EN, Vref.

Figure 1 shows the schematic; Table 4 lists the devices. The core has three branches: left (M7, M5, RB, M3, M1, Q1), middle (M8, M6, RA, M4, M2, R1, Q2) and output (M9, M10, R2, Q3). The PMOS mirror (M7, M8) and the NMOS mirror (M1, M2) force equal currents into the left and middle branches. Q2 has $N = 8$ unit transistors and Q1 one, so the difference of their emitter–base voltages appears across R1:

$$I = \frac{\Delta V_{EB}}{R_1} = \frac{n_f V_T \ln N}{R_1}, \tag{1}$$

with $V_T = kT/q$ and the ideality factor $n_f \approx 1.01$. M9 and M10 copy this PTAT current into R2 and Q3:

$$V_{REF} = V_{EB3} + I R_2 = V_{EB3} + \frac{R_2}{R_1}\, n_f V_T \ln N. \tag{2}$$

V_{EB3} falls by about 2.06 mV/°C; R_2/R_1 is chosen so that the two slopes cancel. Only resistor ratios appear, so the resistor spread cancels to first order. R1 and Q2 sit on the non-diode side of the NMOS mirror, which makes the operating point stable. RA and RB generate the cascode gate voltages from the branch current ($V_{BN2} - V_{BN1} = I R_B$,

$V_{BP1} - V_{BP2} = IR_A$), and the cascodes raise the output resistance of the mirrors, which improves the line sensitivity and the PSRR. The loop also has a zero-current state, which the start-up circuit removes.

2.2 Devices and sizes

Table 4: Devices of the bandgap reference (3.3 V devices; PMOS bulks at VDD, NMOS bulks at GND).

Device	Type	W / L or value	Role
M1, M2	`nmos3v_mac`	8 / 4 µm	NMOS mirror
M3, M4	`nmos3v_mac`	16 / 4 µm	NMOS cascodes
M7, M8, M9	`pmos3v_mac`	16 / 4 µm	PMOS mirror and output copy
M5, M6, M10	`pmos3v_mac`	32 / 4 µm	PMOS cascodes
R1	`rphripoly`	107 kΩ (L 201.955 µm)	sets the PTAT current
RA, RB	`rphripoly`	450 kΩ (L 849.715 µm)	cascode self-bias
R2	`rphripoly`	1.252 MΩ (3 × 788.025 µm)	output resistor
Q1, Q3	`vpnp_mis`	10 × 10 µm, m = 1	unit vertical PNP
Q2	`vpnp_mis`	10 × 10 µm, m = 8	8× vertical PNP
MSU	`nmosnvt3v_mac`	2 / 4 µm	native start-up device
MSW	`nmos3v_mac`	1 / 1 µm	start-up path switch
MPU1, MPU2	`pmos3v_mac`	2 / 1 µm	pull-ups of VBP1, VBP2
MPD1, MPD2	`nmos3v_mac`	1 / 1 µm	pull-downs of VBN1, VBN2
MINVP, MINVN	`pmos3v_mac`, `nmos3v_mac`	2 / 0.35, 1 / 0.35 µm	EN → ENB inverter

The `_mac`/`_mis` models (nch3_mac, pch3_mac, nanch3_mac, pnp10_mis) carry the mismatch statistics needed for Monte Carlo. The resistors are high-resistance poly (1.037 kΩ/□, W = 2 µm); R2 exceeds the 1 mm pcell limit and uses three segments. Q1 and Q2 form a 3×3 common-centroid group with Q1 in the centre.

2.3 Start-up and enable

The start-up is a single native NMOS, MSU ($V_{th} \approx -67$ mV), with its gate at VBN1, its source at VBN2 (across RB) and its drain at VBP1 through the switch MSW, so

$V_{GS,MSU} = -IR_B$. In the zero-current state $V_{GS} = 0 > V_{th}$ and MSU conducts: it pulls VBP1 (and through RA also VBP2) down and drives current through RB into VBN1, which turns on both mirrors. Gate and source rise together, so the kick does not stall like a source follower. Once 510 nA flows, $V_{GS,MSU} = -229$ mV and MSU leaks only 80 pA. One device is enough: without current, RA and RB tie VBP2 to VBP1 and VBN2 to VBN1, so there is only one loop to start.

With EN = 0, MPU1/MPU2 pull VBP1/VBP2 to VDD and MPD1/MPD2 pull VBN1/VBN2 to GND, and MSW blocks the path through MSU. The supply current is then 42 pA and $V_{REF} \approx 0.32$ V (the emitter–base voltage of Q3 at pA leakage). ENB comes from an inverter inside the cell, so the symbol has a single EN pin.

3 Design Trade-offs

Table 5: Main design trade-offs.

Decision	Gain	Cost
Branch current 0.5 µA	Power 5.05 µW typical, at most 8.977 µW at any corner	Large resistors (R2 = 1.252 MΩ)
Long transistors ($L = 4$ µm)	Low mismatch: $\sigma(V_{REF}) = 17.56$ mV, accuracy ±4.8 %	Larger devices
Cascodes with self-bias (RA, RB)	PSRR 69.9 dB at 1 Hz, line sensitivity 678.6 µV/V	Higher minimum supply, 2.09 V (left branch)

The branch current is set by the power limit: the worst corner (FF, ff_res, 3.63 V, 125 °C) uses 8.977 µW, about 10 % below 10 µW. The spread of V_{REF} is set by MOS mismatch (Section 8.3), which falls as $1/\sqrt{WL}$. The cascodes cost headroom, but the minimum supply stays at least 0.68 V below 2.97 V at every corner.

4 Design Equations and Hand Calculations

4.1 Branch current, R1 and power

$P < 10$ µW at 3.63 V allows 2.75 µA, i.e. < 0.92 µA per branch. The current rises with temperature ($I \propto T/R_1(T)$: 1.33× from 27 to 125 °C, about 7 % more from the resistor temperature coefficient) and at ff_res (15–20 %), so $I \approx 0.5$ µA per branch was chosen.

With $\Delta V_{EB} = V_T \ln 8 = 53.8\,\mathrm{mV}$:

$$R_1 = \frac{V_T \ln 8}{I} = \frac{53.8\,\mathrm{mV}}{0.5\,\mu\mathrm{A}} \approx 107\,\mathrm{k\Omega}. \tag{3}$$

The simulated ΔV_{EB} is 54.5 mV ($n_f \approx 1.01$), giving 509 nA (simulated 510 nA, 5.05 µW). The worst-case power estimate of 9.2–9.6 µW compares with the simulated 8.977 µW.

4.2 R2 and output voltage

Setting the derivative of Eq. (2) to zero:

$$\frac{R_2}{R_1} = \frac{-\,dV_{EB3}/dT}{n_f (k/q) \ln 8} = \frac{2.06\,\mathrm{mV/K}}{0.18\,\mathrm{mV/K}} \approx 11.4 \quad (R_2 \approx 1.21\,\mathrm{M\Omega}). \tag{4}$$

The V_{EB} slope is steeper than the textbook $-1.8\,\mathrm{mV/K}$ because of the low current density in the 10×10 µm emitter. Sweeping R2 for $V_{REF}(125\,^\circ\mathrm{C}) = V_{REF}(-40\,^\circ\mathrm{C})$ (Section 6.1), which also absorbs the V_{EB} curvature, gives $R_2 = 1.252\,\mathrm{M\Omega}$ ($R_2/R_1 = 11.70$), and

$$V_{REF} = V_{EB3} + IR_2 = 0.617\,\mathrm{V} + 510\,\mathrm{nA} \times 1.252\,\mathrm{M\Omega} \approx 1.255\,\mathrm{V}. \tag{5}$$

4.3 Cascode bias resistors RA and RB

IR must cover the saturation voltage of the main device, the cascode's body effect and a margin: $IR \approx 0.13 + 0.025 + 0.06 \approx 0.2$–$0.25\,\mathrm{V}$, so $R_A = R_B \approx 450\,\mathrm{k\Omega}$ (simulated $IR = 229\,\mathrm{mV}$). Then $V_{DS1} = IR_B - (V_{GS3} - V_{GS1}) \approx 0.204\,\mathrm{V}$ (simulated 0.195 V) and $V_{SD8} = IR_A - (V_{SG6} - V_{SG8}) = 0.181\,\mathrm{V}$. All devices are in weak to moderate inversion ($IC \approx 0.4$–1.1); the simulated saturation margins $|V_{DS}| - |V_{DS,sat}|$ are 134 mV (M1/M2) and 102 mV (M7–M9).

4.4 Minimum supply per branch

Each branch fails when its cascode leaves saturation; all other node voltages are fixed by the loop or referred to VDD (values from the operating point in Section 6.2):

The equivalent form $V_{DD,min} = V_{DD} - (|V_{DS}| - |V_{DS,sat}|)$ of the branch cascode is used at every corner (Table 10). The left branch limits at about 2.09 V; the simulated 1 % drop of V_{REF} is at 1.979 V because weak-inversion devices leave saturation gradually.

Table 6: Minimum supply per branch at tt, 27 °C.

Branch	Limit	Expression and values (V)	$V_{DD,min}$
Left	M5	$V_{EB1} + V_{GS1} + IR_B + V_{SG8} + IR_A - V_{SG5} + \lvert V_{DS,sat5}\rvert$ $= 0.617 + 0.998 + 0.229 + 0.748 + 0.229 - 0.796 + 0.065$	2.09 V
Middle	M4	$V_{SG8} + IR_A + (V_{BN2} - V_{GS4}) + V_{DS,sat4}$ $= 0.748 + 0.229 + (1.844 - 1.032) + 0.051$	1.84 V
Output	M10	$V_{SG8} + IR_A - V_{SG10} + V_{REF} + \lvert V_{DS,sat10}\rvert$ $= 0.748 + 0.229 - 0.796 + 1.255 + 0.065$	1.50 V

4.5 Mismatch estimate

With $A_{VT} \approx 7\,\text{mV·µm}$, $\sigma(\Delta V_{th})$ is 1.24 mV for M1/M2 and 0.88 mV for M7–M9. Scaled to the PTAT term $IR_2 = 0.639\,\text{V}$, the M1/M2 offset gives 14.5 mV, the M8 → M9 copy error 11.2 mV and the M7/M8 copy error 5.3 mV: 19.1 mV in total ($3\sigma \approx 4.6\,\%$), against 17.56 mV from Monte Carlo.

Table 7: Hand calculations versus simulation (tt, 27 °C, 3.3 V unless noted).

Quantity	Hand calculation	Simulation
Branch current	0.5 µA (target)	510 nA
ΔV_{EB}	53.8 mV (ideal)	54.5 mV
R_2/R_1	11.4 (linear)	11.70 (end-point tuning)
V_{REF}	1.255 V	1.255 V
IR_A, IR_B	0.225 V	0.229 V
V_{DS1}	0.204 V	0.195 V
Minimum supply (left branch)	2.09 V	1.979 V (1 % drop)
Worst-case power	9.2–9.6 µW	8.977 µW (PVT)
$\sigma(V_{REF})$	19.1 mV	17.56 mV (Monte Carlo)
V_{REF} in power-down	0.326 V	0.323 V

5 Implementation and Simulation Setup

5.1 Library, cells and testbenches

All cells are in library `JaiSrikar_CMOS` (technology library `tsmc18`, model file `cr018gpii_v1d0.scs`, Figure 2), path:
`/cad/cad_users/IIIT2024702001/JaiSrikar_CMOS`.
The bandgap is cell `mid-project-bgr` (schematic and symbol, pins VDD, GND, EN, Vref) and appears as instance I0 in every testbench. The testbenches are schematic views

of cell `BGR_Test`, set up in ADE Explorer and ADE Assembler (Maestro); design variables `vdd_var` = 3.3, `en` = 1, `R2_Var` = 1.252M.

Table 8: Testbenches (cell `BGR_Test`).

Testbench	Sources	Load	Used for
`TB_BGR_DC`	V0: VDD = vdd_var, AC magnitude 1; V2: EN = en·vdd_var; V1: GND	1 fF	DC, temperature, VDD, PSRR, MC
`TB_BGR_SU`	V0: PWL 0 V until 1 µs, ramp to vdd_var at 11 µs; drives VDD and EN	1 pF	start-up (typical, PVT, MC)
`TB_BGR_SEQ`	V0: VDD ramp as above; V2: EN pulse (delay 50 µs, rise/fall 1 µs, width 250 µs, period 500 µs)	1 fF	power sequencing
`no-su-ver-tb`	core without start-up and enable devices, drawn in the testbench; VDD ramp as above	1 fF	settling without start-up

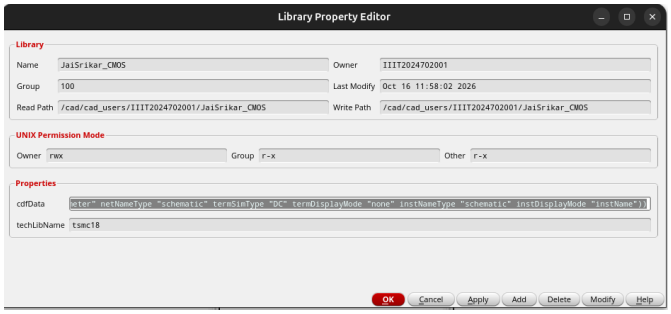
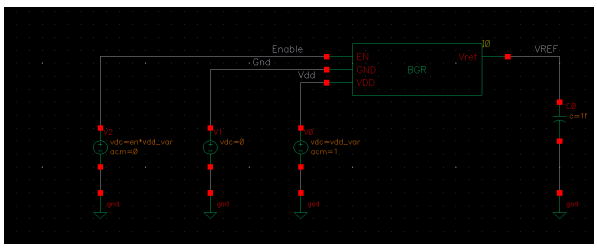
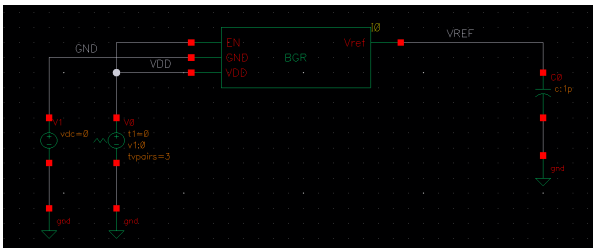


Figure 2: Library `JaiSrikar_CMOS`: path `/cad/cad_users/IIIT2024702001/JaiSrikar_CMOS`, technology library `tsmc18`.

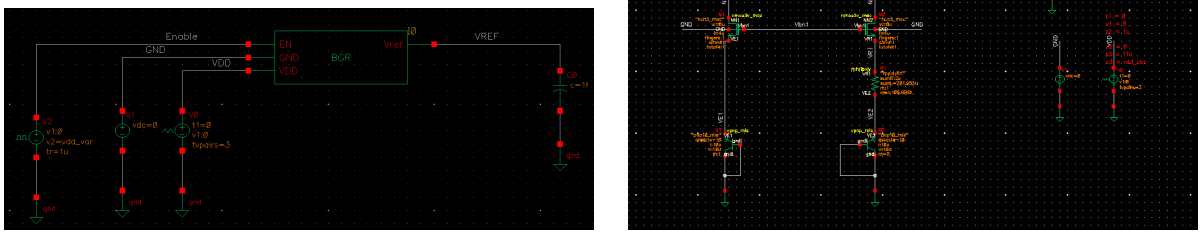


(a) `TB_BGR_DC`: V0 VDD = vdd_var (AC 1 for PSRR), V2 EN = en·vdd_var, C0 = 1 fF.



(b) `TB_BGR_SU`: V0 ramps VDD and EN (tied) from 0 to vdd_var between 1 and 11 µs; C0 = 1 pF.

Figure 3: Testbenches with the bandgap symbol (instance I0, pins EN, GND, VDD, Vref).



(a) `TB_BGR_SEQ`: V0 ramps VDD, V2 pulses EN (delay 50 µs, rise/fall 1 µs, width 250 µs, period 500 µs); C0 = 1 fF.

(b) `no-su-ver-tb`: core without start-up and enable devices drawn in the testbench; VDD ramp 0 to 3.3 V.

Figure 4: Testbenches for power sequencing and for the test without start-up circuit.

5.2 Analyses

Table 9: Analyses.

Test	Analysis
DC operating point	dc with Save DC Operating Point; en = 1 (running) and en = 0 (power-down)
R2 tuning	dc temperature sweep −40 to 125 °C (step 1), R2_Var 1.2–1.3 MΩ in 2 kΩ steps
Temperature	dc temperature sweep −40 to 125 °C, step 1 °C
Supply	dc sweep of vdd_var from 0 to 4 V, 400 steps
PSRR	ac 1 Hz to 1 GHz, logarithmic, 20 points/decade, AC source 1 on VDD
Start-up, sequencing, no start-up	tran 0 to 1 ms, conservative, no initial conditions or nodesets

With the start-up circuit in place the DC solution is the operating state, so no nodesets are needed.

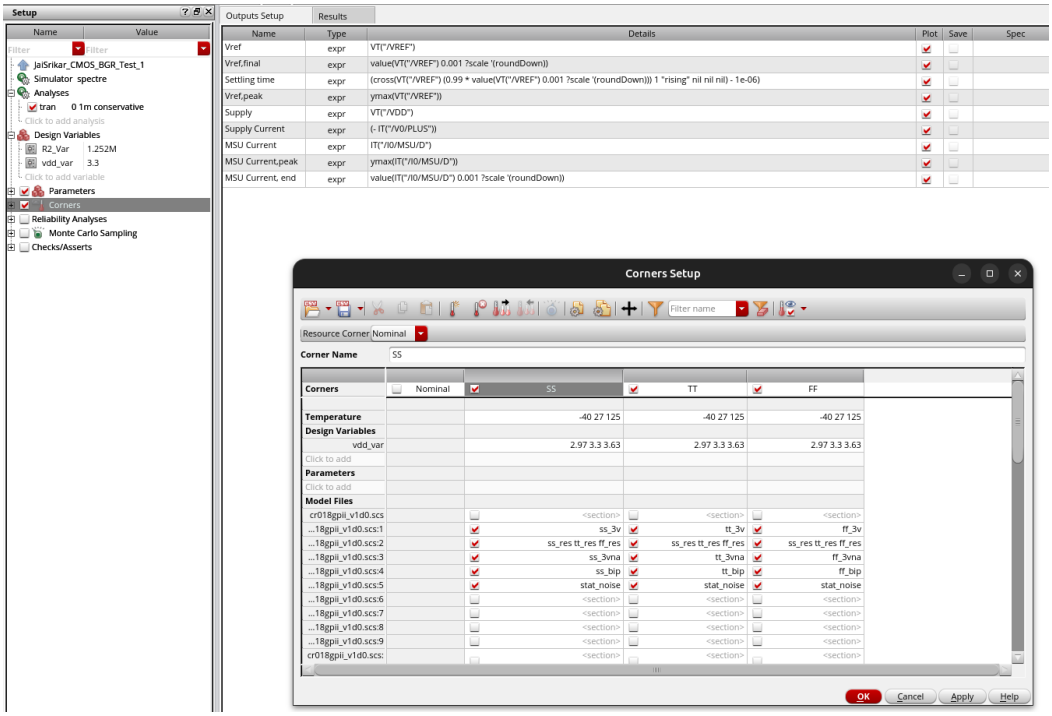
5.3 Output expressions

Table 10: Maestro output expressions.

Output	Expression
Temperature views	
VREF at T	`value(VS("/VREF") T), T = -40, 27, 125`
CTAT, PTAT	`VS("/IO/VE3"); VS("/VREF") - VS("/IO/VE3")`
VREF difference	`value(VS("/VREF") 125) - value(VS("/VREF") -40)`
TC (ppm/°C)	`((value(VS("/VREF") 125) - value(VS("/VREF") -40)) / (value(VS("/VREF") 27)*165))*1e6`
Slope at 27 °C / average	`value(deriv(X) 27); (value(X 125) - value(X -40))/165, X = CTAT or PTAT`
Supply current	`-IS("/VO/PLUS")`
Max supply current, power	`max(ymax(clip(-IS("/VO/PLUS") -31 125)) ymax(clip(-IS("/VO/PLUS") -40 -37))); times VAR("vdd_var")`
I(MSU) at 125 °C	`value(IS("/IO/MSU/D") 125)`
Supply views	
Line sensitivity	`(value(VS("/VREF") 3.63) - value(VS("/VREF") 2.97))/0.66`
VDD,min (1 % drop)	`cross(VS("/VREF") 0.99*value(VS("/VREF") 3.3) 1 "rising")`
DC operating point over corners	
VDD,min left / middle / output	`VAR("vdd_var") + OP("/IO/M5" "vds") - OP("/IO/M5" "vdsat"); VAR("vdd_var") - OP("/IO/M4" "vds") + OP("/IO/M4" "vdsat"); VAR("vdd_var") + OP("/IO/M10" "vds") - OP("/IO/M10" "vdsat")`
Supply power	`VAR("vdd_var")*(-IDC("/VO/PLUS"))`
PSRR views	
PSRR	`-dB20(VF("/VREF")); value(PSRR f), f = 1, 1k, 100k`
f(40 dB)	`cross(PSRR 40 1 "falling")`
Start-up views	
VREF final, peak	`value(VT("/VREF") 1m); ymax(VT("/VREF"))`
Settling time	`cross(VT("/VREF") 0.99*value(VT("/VREF") 1m) 1 "rising") - 1u`
I(MSU) peak, end	`ymax(IT("/IO/MSU/D")); value(IT("/IO/MSU/D") 1m)`
Supply power (final)	`VAR("vdd_var")*value(-IT("/VO/PLUS") 1m)`

The clip windows skip the MOS-model glitch between −36 and −31 °C, which is ignored as instructed.

5.4 Corner and Monte Carlo set-up



(a) PVT corners.



(b) Monte Carlo corner MC_TT.

Figure 5: Corner set-up. (a) Corner set-up of the PVT runs (model file `cr018gpii_v1d0.scs`): process columns SS/TT/FF with the MOS (ss/tt/ff_3v), native (ss/tt/ff_3vna) and BJT (ss/tt/ff_bip) sections tied; inside each column the resistor corners ss/tt/ff_res, VDD 2.97/3.3/3.63 V and temperature −40/27/125 °C (81 corners). A variable swept by the analysis is left empty, which gives 27 corners. (b) Monte Carlo corner MC_TT: statistical sections of `cr018gpii_v1d0.scs` and VDD = 3.3 V. Sampling: 200 random points, statistical variation "All" (process and mismatch), or "Mismatch" with an instance selection for the mismatch-only runs.

5.5 Views and history names

Each run has its own Maestro view in cell `BGR_Test` (test `JaiSrikar_CMOS_BGR_Test_1`); the results are saved under these history names:

Table 11: Maestro views and history names.

Run	TB	Tool	View	History
R2 tuning (tt)	DC	Explorer	`R2_Tune_Validation`	`Temp_sweep_R2`
DC operating point, EN = 1 / 0 (tt)	DC	Assembler	`DC-OP`	`EN_High`, `EN_Low`
Temperature sweep (tt)	DC	Explorer	`Temp_Sweep`	`Temp_sweep_nom`
Supply sweep (tt)	DC	Explorer	`VDD_Sweep`	`VDD_Sweep+LS`
PSRR (tt)	DC	Explorer	`AC_Analysis`	`PSRR`
Start-up (tt)	SU	Explorer	`SU_Test`	`SU_test`
Power sequencing (tt)	SEQ	Explorer	`SEQ_Test`	`SEQ_Test`
No start-up (tt)	no-su-ver-tb	Explorer	`No_SU-test`	`No_SU`
Start-up, 81 corners	SU	Explorer	`SU_PVT`	`SU_PVT-only`
DC operating point, 27 corners	DC	Explorer	`DC_PVT`	`DC_PVT-only`
Temperature sweep, 27 corners	DC	Explorer	`Temp_Sweep_PVT`	`TEMP_PVT`
Supply sweep, 27 corners	DC	Explorer	`VDD_PVT`	`VDD_PVT`
PSRR, 81 corners	DC	Assembler	`AC_PVT`	`PSRR_PVT`
Monte Carlo, temperature (200)	DC	Assembler	`Temp_Sweep_MC`	`Temp_MC`
Monte Carlo, start-up (200)	SU	Assembler	`SU_MC`	`SU_MC_ONLY`
Mismatch only, 4×200	DC	Explorer	`mismatch`	`BJT-only-mismatch`, `RES-only-mismatch`, `NMOS-only-mismatch`, `PMOS-only-mismatch`

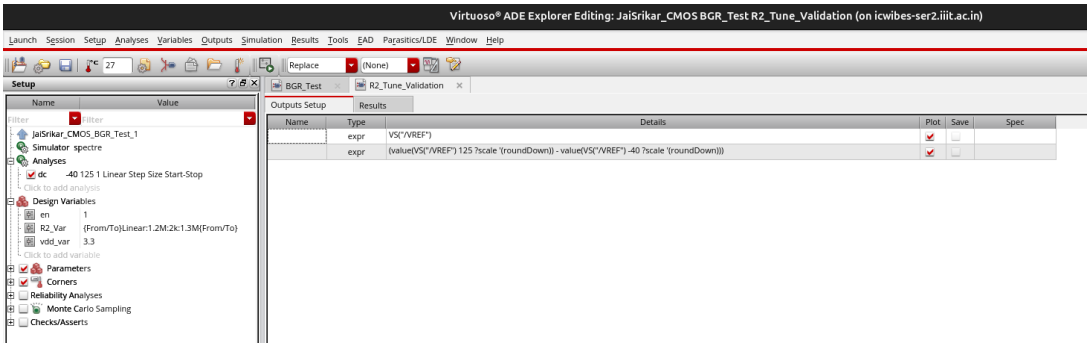
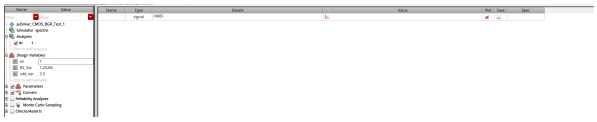


Figure 6: Set-up of the R2 tuning run (view `R2_Tune_Validation`): dc temperature sweep −40...125 °C with R2_Var swept from 1.2 to 1.3 MΩ in 2 kΩ steps.



(a) DC analysis with Save DC Operating Point; design variables en, R2_Var, vdd_var.



(b) Saved histories `EN_High` and `EN_Low`.

Figure 7: Set-up of the DC operating-point view `DC-OP`.

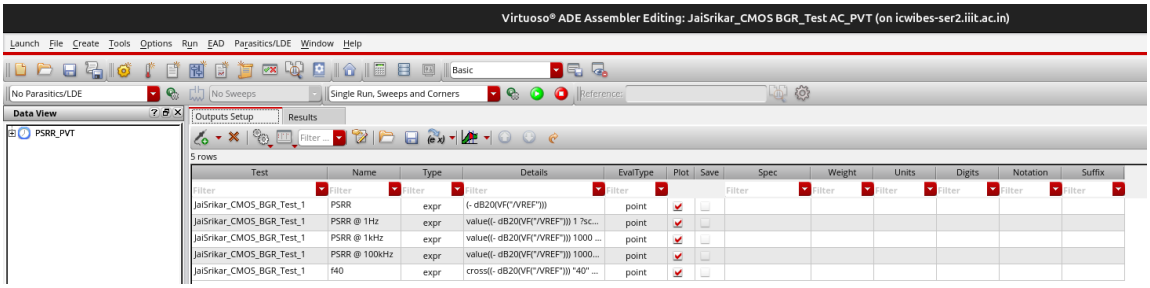


Figure 8: Output set-up of view `AC_PVT` (PSRR).

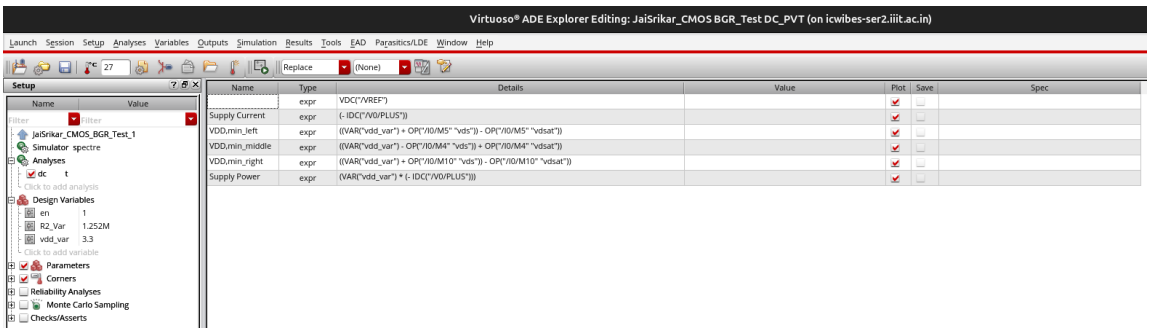


Figure 9: Output set-up of view `DC_PVT` (minimum supply per branch).

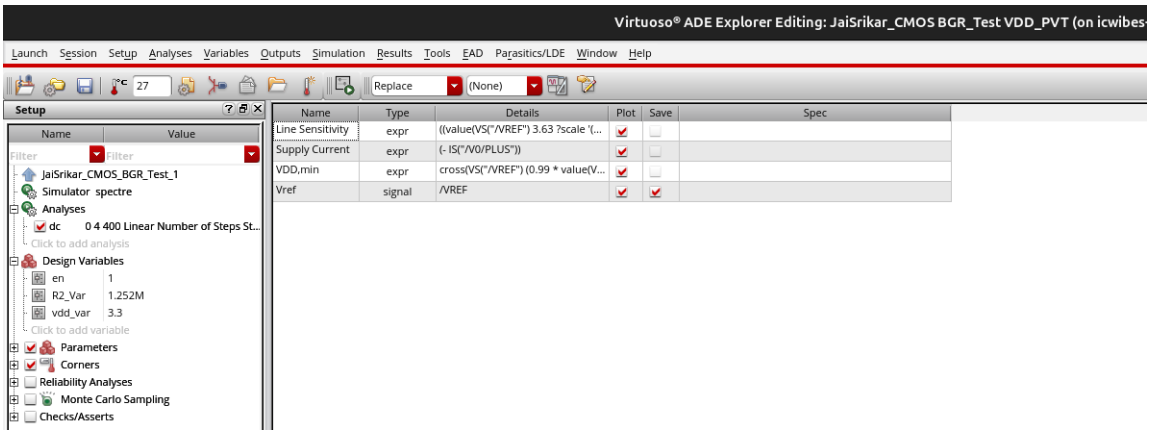


Figure 10: Output set-up of view `VDD_PVT` (line sensitivity and minimum supply).

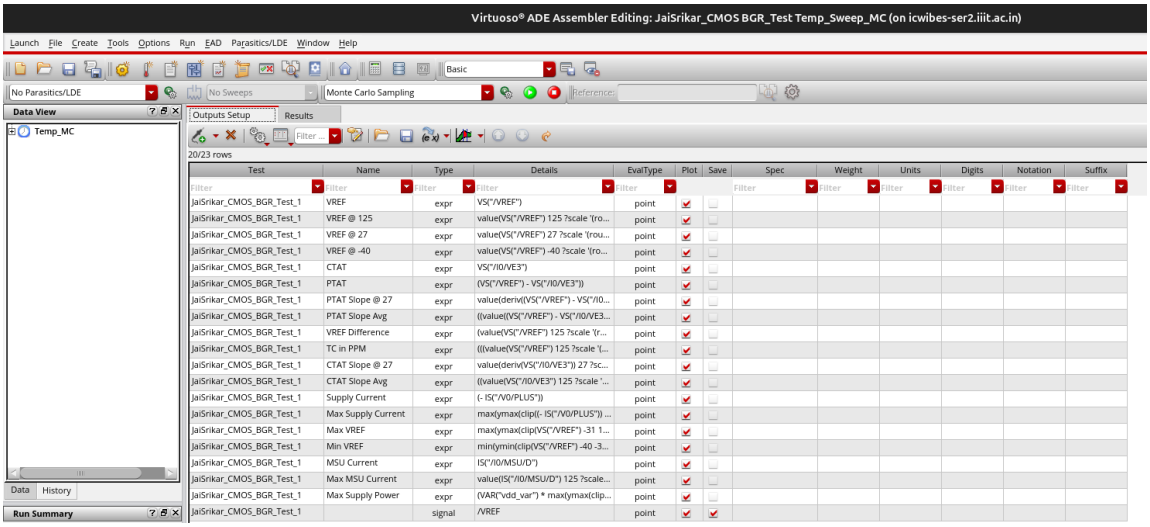


Figure 11: Output set-up of view `Temp_Sweep_MC` (temperature outputs; same outputs in `Temp_Sweep` and `Temp_Sweep_PVT`).

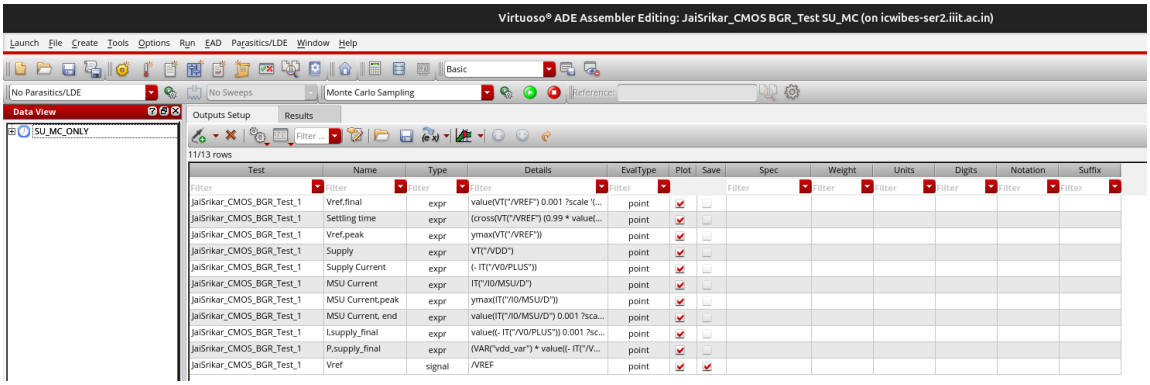
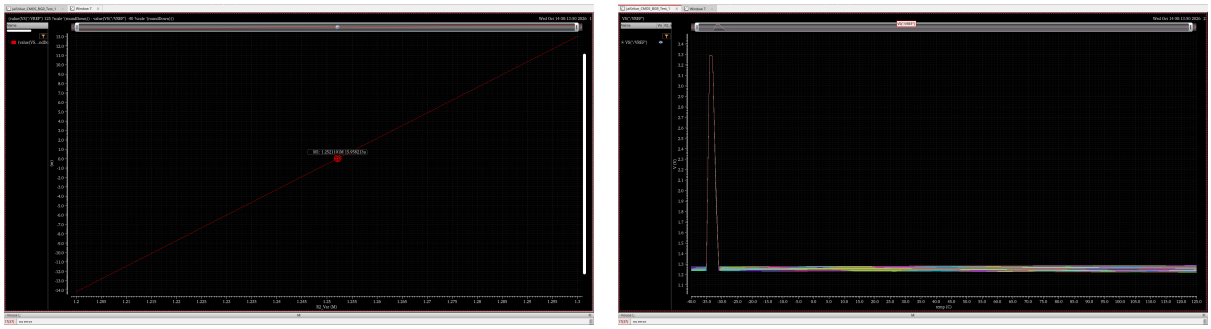


Figure 12: Output set-up of view `SU_MC` (start-up outputs; same outputs in `SU_Test` and `SU_PVT`).

6 Typical Results (tt, 27 °C, 3.3 V)

6.1 R2 tuning

History `Temp_sweep_R2`. The end-point difference crosses zero at 1.2521 MΩ (Figure 13a), so $R_2 = 1.252\,\mathrm{M\Omega}$; it changes by 0.27 mV per kΩ (expected 0.28 mV/kΩ).



(a) End-point difference $V_{REF}(125\,^\circ\mathrm{C}) - V_{REF}(-40\,^\circ\mathrm{C})$ versus R2; zero crossing at 1.2521 MΩ.

(b) V_{REF} versus temperature for R2 from 1.2 to 1.3 MΩ.

Figure 13: R2 tuning (tt, 3.3 V): the zero crossing sets R2 = 1.252 MΩ. History `Temp_sweep_R2`.

6.2 DC operating point and power-down

History `EN_High` (EN = 1): every branch carries about 510 nA and all core transistors are saturated (Figure 14). MSU is off at 80 pA (Figure 15a), and the supply current is 1.530 µA, i.e. 5.05 µW (Figure 16a). History `EN_Low` (EN = 0): all core devices are off, the switches are on and MSW blocks the start-up path (Figure 15b); the supply current is 42 pA (Figure 16b) and $V_{REF} = 0.323\,\mathrm{V}$ (Figure 17; hand estimate 0.326 V).

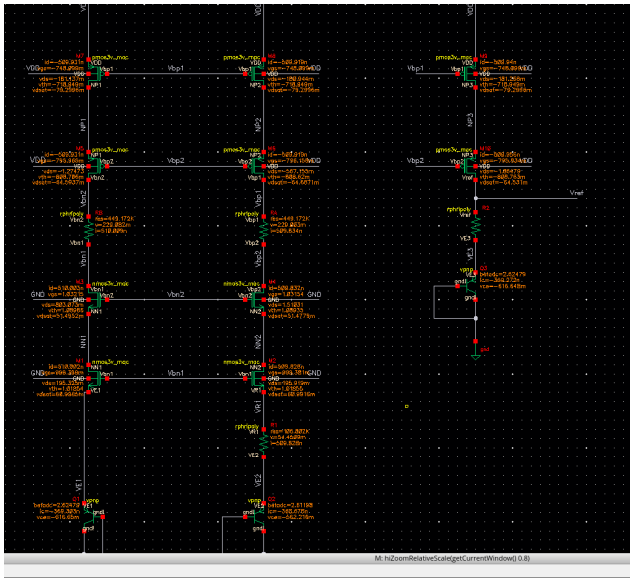
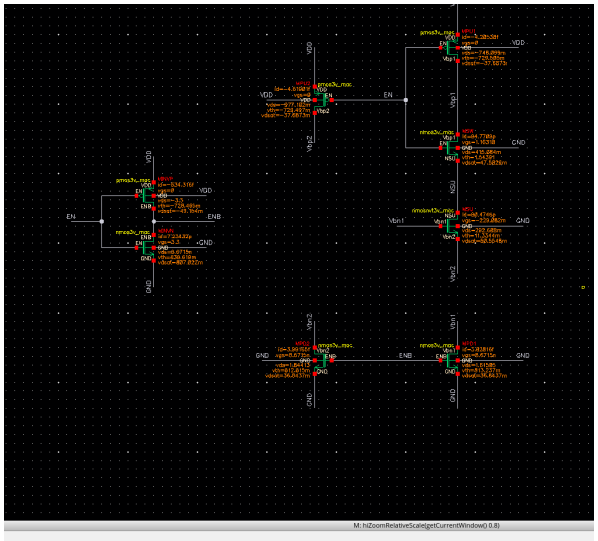
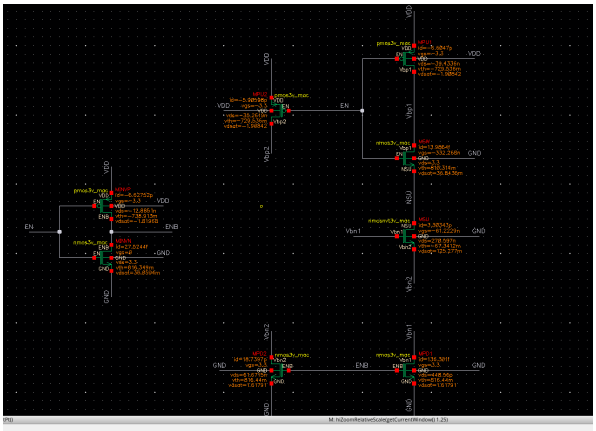


Figure 14: DC operating point with EN = 1, bandgap core annotated: all branches carry ≈ 510 nA. History `EN_High`.

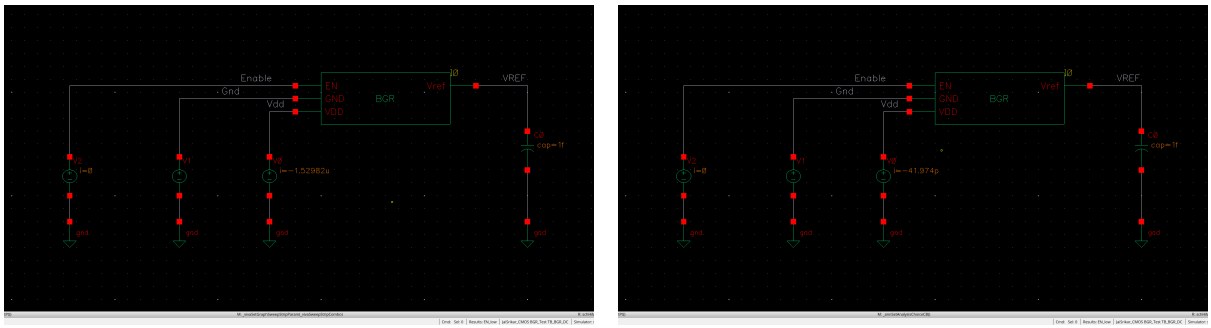


(a) EN = 1 (`EN_High`): MSU off (80 pA, V_{GS} = −229 mV); pull-ups and pull-downs off.



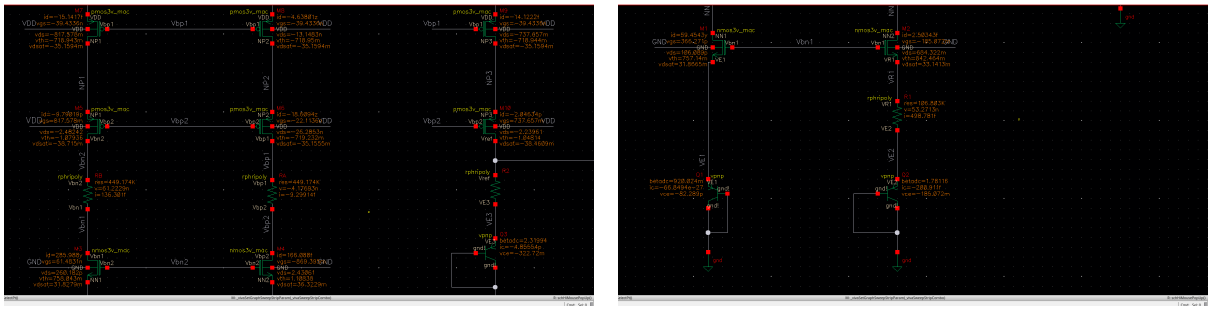
(b) EN = 0 (`EN_Low`): pull-ups and pull-downs on; MSW blocks the start-up path (MSU 3.3 pA, MSW 14 fA).

Figure 15: DC operating point of the start-up and enable devices.



(a) EN = 1: 1.530 µA, i.e. 5.05 µW (`EN_High`). **(b)** EN = 0 (power-down): 42 pA (`EN_Low`).

Figure 16: Testbench annotated with the supply current.



(a) Upper part of the core. **(b)** Lower part of the core (M1, M2, R1, Q1, Q2).

Figure 17: DC operating point with EN = 0 (power-down): all devices off; $V_{REF} = V_{EB3} \approx 0.323$ V. History `EN_Low`.

6.3 Temperature sweep

History `Temp_sweep_nom` (Figures 18 and 19): $V_{REF} = 1.255$ V; $V_{REF}(125) - V_{REF}(-40) = -14$ µV, i.e. $TC = -0.068$ ppm/°C; CTAT and PTAT slopes ∓ 2.06 mV/°C at 27 °C; maximum supply current 2.17 µA (7.17 µW) at 125 °C; I(MSU) 80 pA at 27 °C and 485 pA at 125 °C.

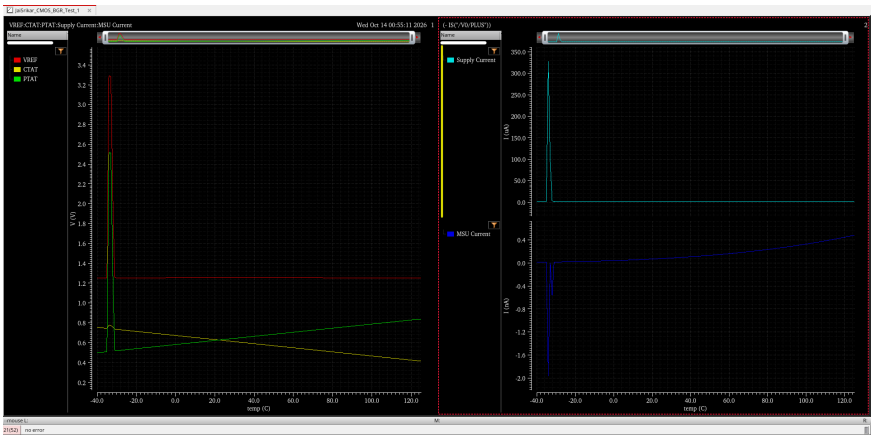


Figure 18: Temperature sweep −40...125 °C (tt, 3.3 V): V_{REF}, V_{EB} of Q3 (CTAT), $V_{REF} - V_{EB3}$ (PTAT), supply current and start-up current. The spike between −36 and −31 °C is a known MOS-model artefact and is ignored. History `Temp_sweep_nom`.

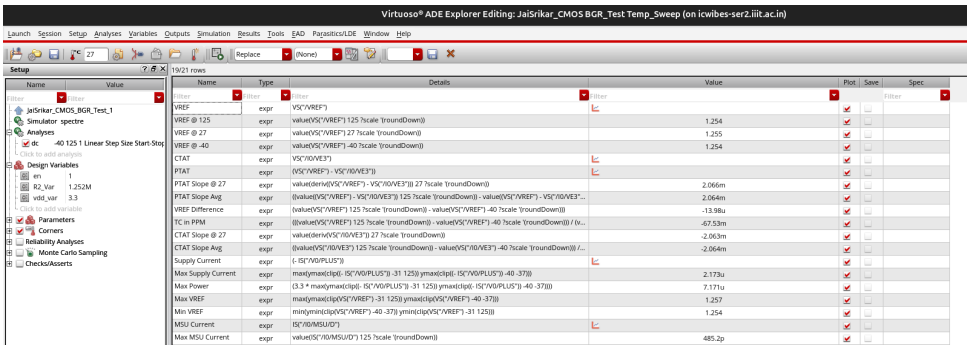


Figure 19: Outputs of the typical temperature sweep: V_{REF} at −40/27/125 °C, end-point TC, CTAT and PTAT slopes, maximum supply current and power, maximum and minimum V_{REF}. History `Temp_sweep_nom`.

6.4 Supply sweep and line sensitivity

History `VDD_Sweep+LS` (Figures 20 and 21). Near 1.1 V V_{REF} steps up where the start-up acts; above about 2.05 V it is flat. Line sensitivity over 2.97–3.63 V: 678.6 µV/V (0.45 mV over the range); 1 % drop point 1.979 V.

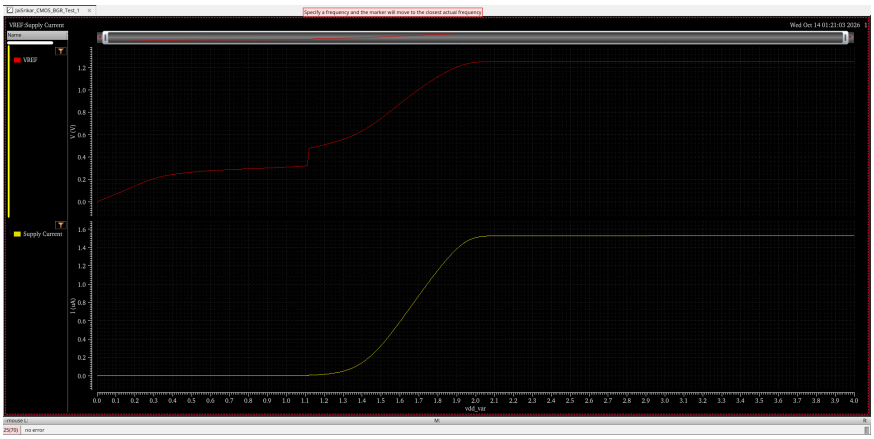


Figure 20: V_{REF} and supply current versus VDD from 0 to 4 V (tt, 27 °C). History `VDD_Sweep+LS`.



Figure 21: Outputs of the supply sweep: line sensitivity over 2.97–3.63 V (678.6 µV/V) and minimum supply, V_{REF} 1 % below its 3.3 V value (1.979 V). History `VDD_Sweep+LS`.

6.5 PSRR

History `PSRR` (Figures 22 and 23): 69.9 dB at 1 Hz, 58.8 dB at 1 kHz, 21.4 dB at 100 kHz; 40 dB at 9.12 kHz. The DC supply rejection implied by the line sensitivity, 63.4 dB, is of the same order as the 69.9 dB at 1 Hz.

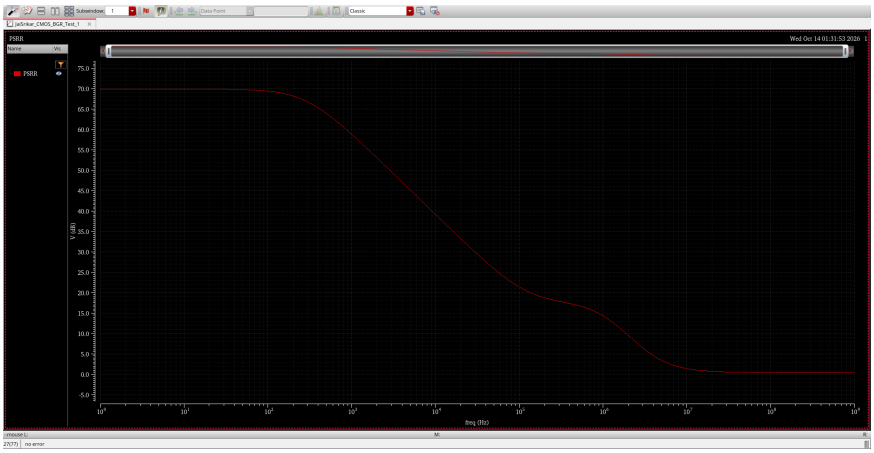


Figure 22: PSRR versus frequency, 1 Hz–1 GHz (tt, 27 °C, 3.3 V, 1 fF load). History `PSRR`.

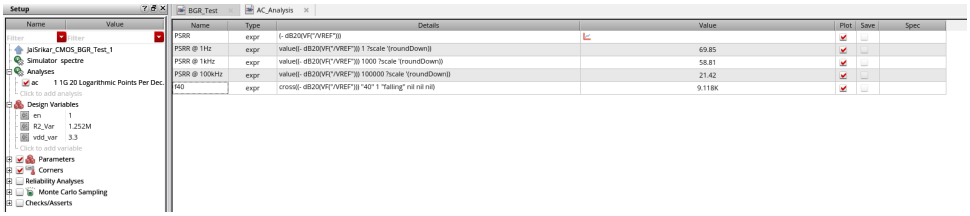


Figure 23: PSRR outputs: 69.9 dB at 1 Hz, 58.8 dB at 1 kHz, 21.4 dB at 100 kHz; 40 dB at 9.12 kHz. History `PSRR`.

6.6 Start-up

History `SU_test` (EN tied to VDD, 1 pF load; Figures 24 and 25). V_{REF} settles to 1.255 V without overshoot, reaching 99 % 16.11 µs after the start of the VDD ramp. I(MSU) peaks at 97.5 nA and falls to 82 pA.



Figure 24: Start-up with EN tied to VDD (tt, 27 °C, 1 pF load): VDD ramp 0 → 3.3 V in 10 µs, V_{REF}, supply current and start-up current I(MSU). History `SU_test`.

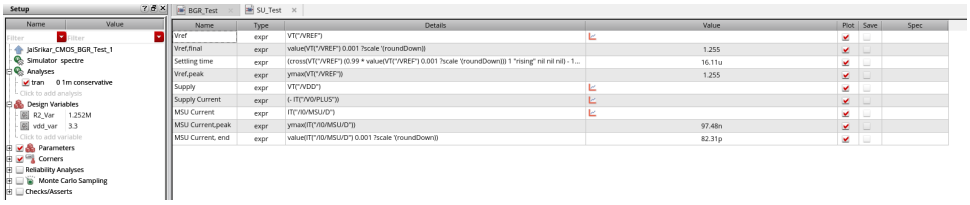


Figure 25: Outputs of the start-up test: final V_{REF} 1.255 V, settling to 99 % in 16.11 µs from the start of the VDD ramp, I(MSU) peak 97.5 nA and 82 pA at 1 ms. History `SU_test`.

6.7 Power sequencing (Deliverable 5)

History `SEQ_Test` (Figures 26 and 27). VDD ramps up while EN is low; EN is then pulsed high at 50 µs and 550 µs. The core starts on both pulses (V_{REF} = 1.255 V) and shuts down whenever EN is low; the short supply-current spikes at the EN edges come from the inverter and the switch gate charge.

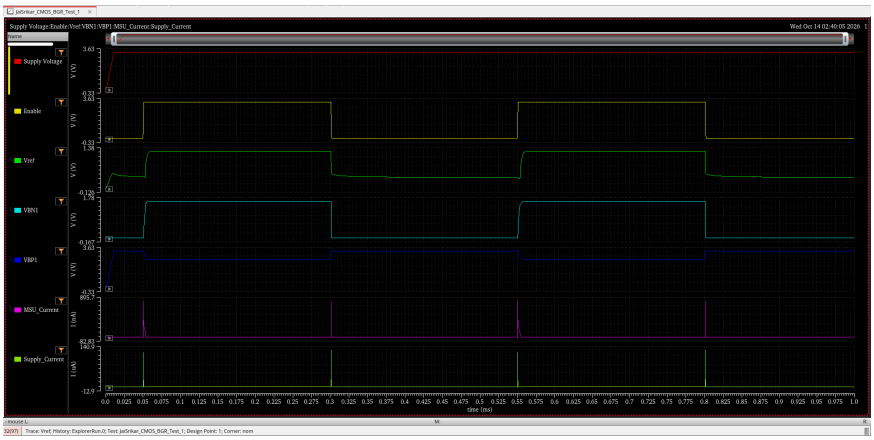


Figure 26: Power sequencing (tt, 27 °C): VDD ramps first, then EN is pulsed high at 50 µs and 550 µs; V_{REF}, VBN1, VBP1, I(MSU) and supply current. The core starts on both pulses and shuts down while EN is low. History `SEQ_Test`.

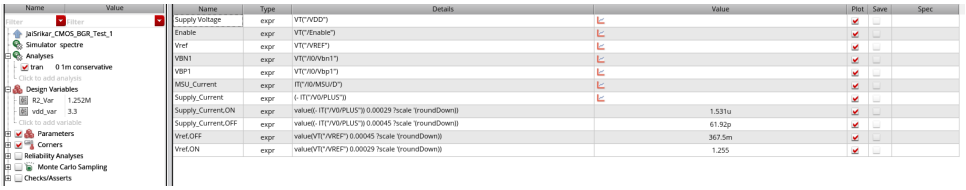
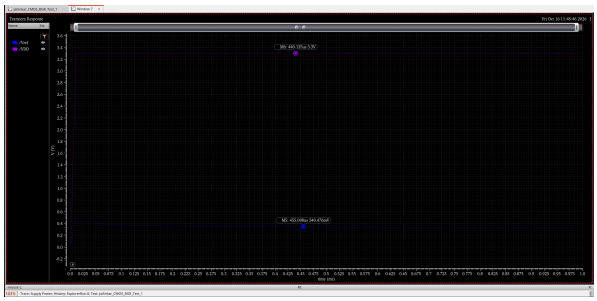


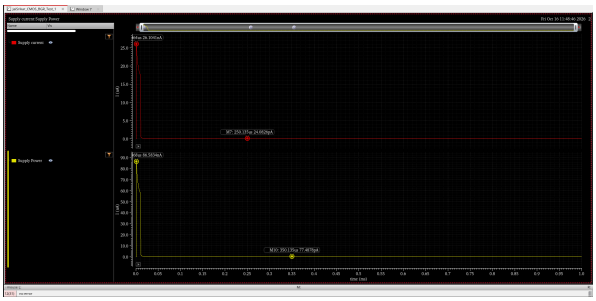
Figure 27: Power-sequencing outputs with EN high and EN low. History `SEQ_Test`.

6.8 Without start-up circuit (Deliverable 21)

History `No_SU` (VDD ramp only, no start-up or enable devices; Figures 28 and 29). V_{REF} peaks at about 0.51 V during the ramp and then stays at about 0.33 V with 24 pA supply current: the core remains in its zero-current state and never comes within 99 % of 1.255 V, so no settling time exists.



(a) V_{REF} and VDD: V_{REF} peaks at $\approx 0.51\,\mathrm{V}$, then stays at $\approx 0.33\,\mathrm{V}$.



(b) Supply current and power: $\approx 26\,\mathrm{nA}$ during the ramp, then $24\,\mathrm{pA}$ (power axis labelled in A; values in W).

Figure 28: VDD ramp without a start-up circuit (tt, 27 °C): the core stays in the zero-current state and never settles to 1.255 V. History `No_SU`.

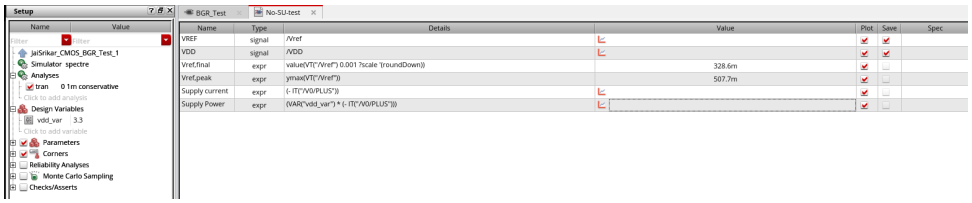


Figure 29: Outputs of the no-start-up test: final V_{REF} 0.329 V, peak 0.508 V. History `No_SU`.

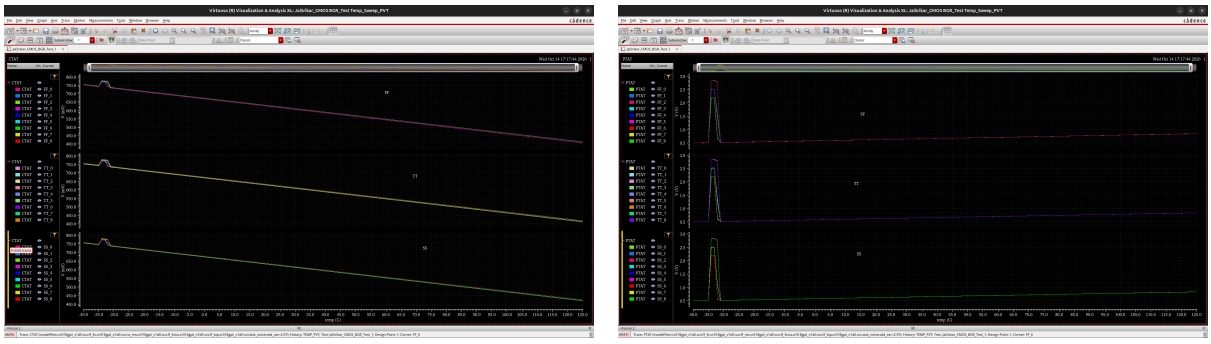
7 PVT Results

7.1 V_{EB}, PTAT voltage and slopes (Deliverables 1–3)

History `TEMP_PVT`, 27 corners (Figure 30; slope expressions in Table 10). The CTAT slope is steepest in FF and shallowest in SS, and the PTAT slope follows it, so the two nearly cancel at every corner (TC −6.0...25.9 ppm/°C).

Table 12: CTAT and PTAT slopes over 27 corners (history `TEMP_PVT`).

Slope (mV/°C)	at 27 °C	average −40...125 °C
CTAT (V_{EB3})	−2.102 ... −2.013	−2.102 ... −2.014
PTAT ($V_{REF} - V_{EB3}$)	2.049 ... 2.120	2.044 ... 2.123



(a) V_{EB} of Q3 (CTAT). **(b)** PTAT voltage $(\Delta V_{EB}/R_1)\cdot R_2 = V_{REF} - V_{EB3}$.

Figure 30: CTAT and PTAT voltages versus temperature, 27 corners grouped by process corner (9 traces each: 3 resistor corners × 3 supplies). History `TEMP_PVT`.

7.2 Supply current versus temperature (Deliverable 4)

The maximum is 2.473 µA, i.e. 8.977 µW at 3.63 V, at FF, ff_res, 125 °C (Figure 31); over all 81 start-up corners the power is 2.99–8.977 µW.

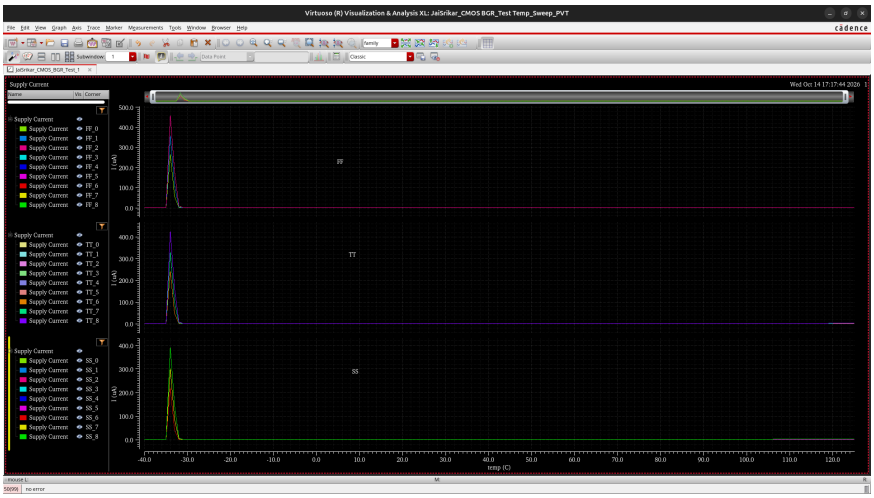


Figure 31: Total supply current versus temperature, 27 corners grouped by process corner; maximum 2.473 µA (8.977 µW at 3.63 V) at FF, ff_res, 125 °C. History `TEMP_PVT`.

7.3 Output voltage versus temperature (Deliverables 6, 19)

V_{REF} follows Eq. (2): the resistor corners cancel in R_2/R_1, so it moves mainly with V_{EB3} and, at FF and 125 °C, with the start-up leakage (Figures 32 and 33).

The "Max VREF" of 1.292 V in Figure 33 occurs only at the 3.63 V corners and contains the tail of the model glitch; outside it the maximum is 1.263 V at 125 °C.

Table 13: V_{REF} over PVT.

Quantity	Range
V_{REF} at -40 / 27 / 125 °C (27 corners, `TEMP_PVT`)	1.250–1.259 / 1.251–1.261 / 1.249–1.263 V
V_{REF} after start-up (81 corners, `SU_PVT-only`)	1.249–1.263 V, $\sigma = 3.26$ mV
$V_{REF}(125) - V_{REF}(-40)$; end-point TC	-1.23 … $+5.37$ mV; -6.0 … 25.9 ppm/°C
I(MSU) at 125 °C	≤ 2.504 nA (FF)

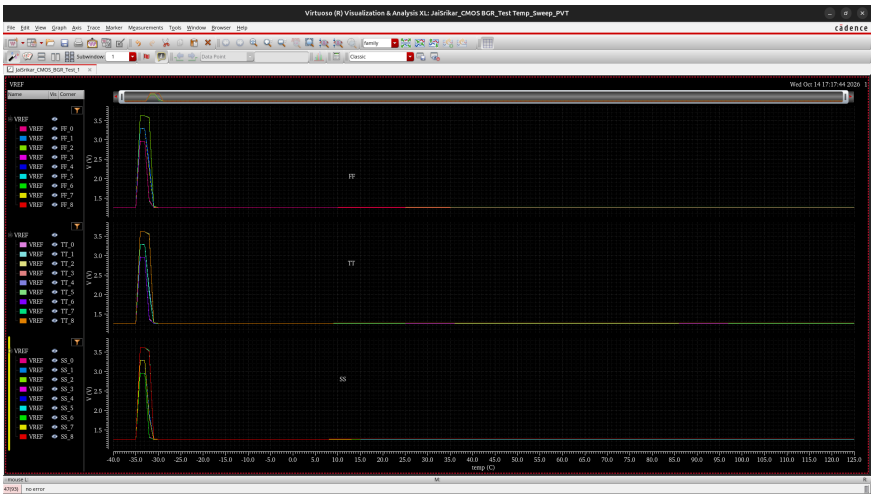
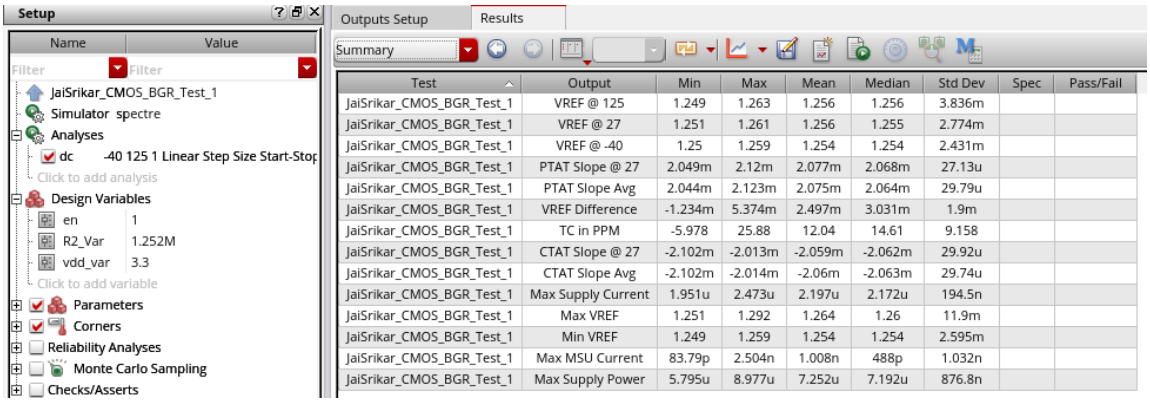


Figure 32: V_{REF} versus temperature, 27 corners grouped by process corner. History `TEMP_PVT`.



Test	Output	Min	Max	Mean	Median	Std Dev	Spec	Pass/Fail
JaiSrikar_CMOS_BGR_Test_1	VREF @ 125	1.249	1.263	1.256	1.256	3.836m		
JaiSrikar_CMOS_BGR_Test_1	VREF @ 27	1.251	1.261	1.256	1.255	2.774m		
JaiSrikar_CMOS_BGR_Test_1	VREF @ -40	1.25	1.259	1.254	1.254	2.431m		
JaiSrikar_CMOS_BGR_Test_1	PTAT Slope @ 27	2.049m	2.12m	2.077m	2.068m	27.13u		
JaiSrikar_CMOS_BGR_Test_1	PTAT Slope Avg	2.044m	2.123m	2.075m	2.064m	29.79u		
JaiSrikar_CMOS_BGR_Test_1	VREF Difference	-1.234m	5.374m	2.497m	3.031m	1.9m		
JaiSrikar_CMOS_BGR_Test_1	TC in PPM	-5.978	25.88	12.04	14.61	9.158		
JaiSrikar_CMOS_BGR_Test_1	CTAT Slope @ 27	-2.102m	-2.013m	-2.059m	-2.062m	29.92u		
JaiSrikar_CMOS_BGR_Test_1	CTAT Slope Avg	-2.102m	-2.014m	-2.06m	-2.063m	29.74u		
JaiSrikar_CMOS_BGR_Test_1	Max Supply Current	1.951u	2.473u	2.197u	2.172u	194.5n		
JaiSrikar_CMOS_BGR_Test_1	Max VREF	1.251	1.292	1.264	1.26	11.9m		
JaiSrikar_CMOS_BGR_Test_1	Min VREF	1.249	1.259	1.254	1.254	2.595m		
JaiSrikar_CMOS_BGR_Test_1	Max MSU Current	83.79p	2.504n	1.008n	488p	1.032n		
JaiSrikar_CMOS_BGR_Test_1	Max Supply Power	5.795u	8.977u	7.252u	7.192u	876.8n		

Figure 33: Temperature sweep over 27 corners, summary: V_{REF} at $-40/27/125$ °C, end-point TC, CTAT and PTAT slopes, maximum supply current and power, maximum and minimum V_{REF}, I(MSU) at 125 °C. Max V_{REF} at the nine 3.63 V corners includes the -36…-31 °C model glitch. History `TEMP_PVT`.

7.4 Line sensitivity (Deliverables 8, 9)

History `VDD_PVT`, 27 corners (Figures 34 and 35). All corners reach the plateau by about 2.2 V and stay flat to 4 V. Line sensitivity 0.375–2.505 mV/V (at most 1.65 mV over 2.97–3.63 V); 1 % drop point 1.757–2.207 V.

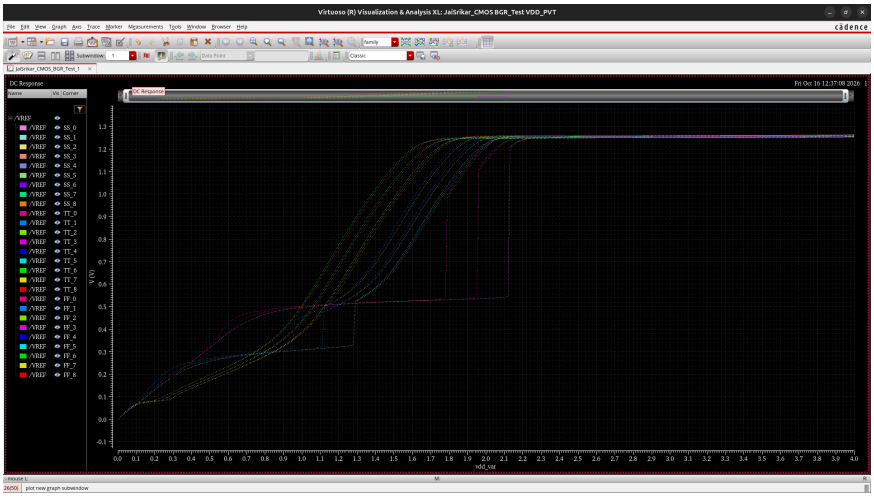


Figure 34: V_{REF} versus VDD from 0 to 4 V over 27 corners (process × resistor × temperature). All corners reach the reference plateau by ≈ 2.2 V and stay flat up to 4 V; the steps between 1.1 and 2.1 V are where the start-up circuit brings the core to its operating point. History `VDD_PVT`.

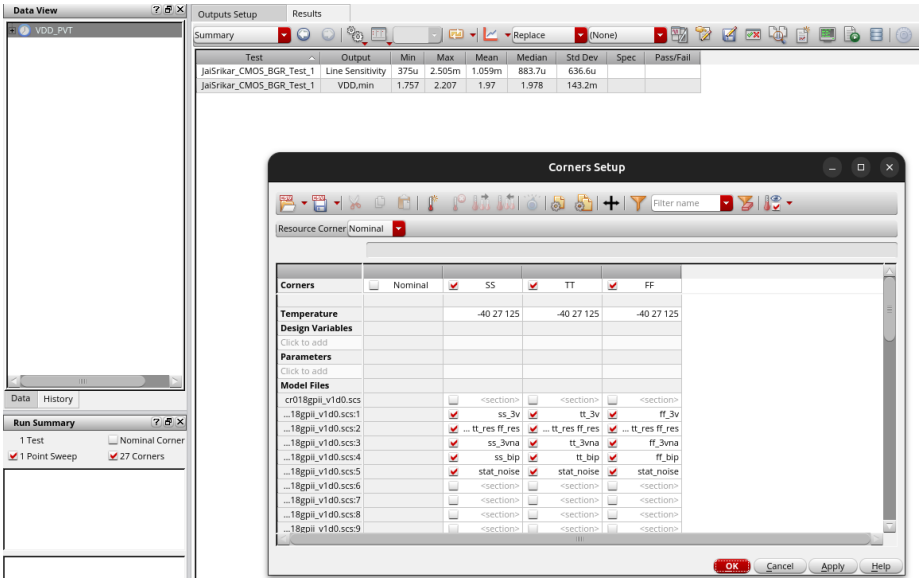


Figure 35: Supply sweep over 27 corners, summary: line sensitivity 0.375–2.505 mV/V, minimum supply 1.757–2.207 V, with the corner set-up. History `VDD_PVT`.

7.5 PSRR (Deliverable 10)

History `PSRR_PVT`, 81 corners (Figures 36 and 37): 47.4–117.3 dB at 1 Hz, 47.2–61.3 dB at 1 kHz, 19.7–23.7 dB at 100 kHz; 40 dB frequency 6.66–12.62 kHz. The low-frequency PSRR exceeds 40 dB at every corner.

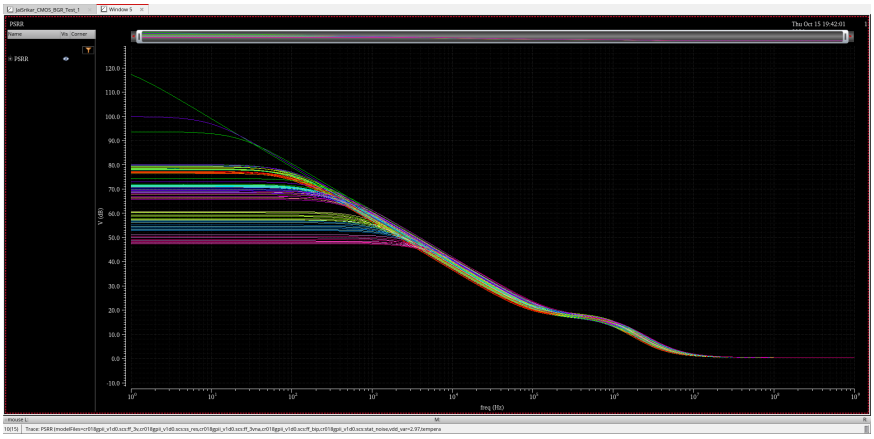


Figure 36: PSRR versus frequency over all 81 PVT corners (VDD 2.97/3.3/3.63 V, −40/27/125 °C, process and resistor corners). History `PSRR_PVT`.

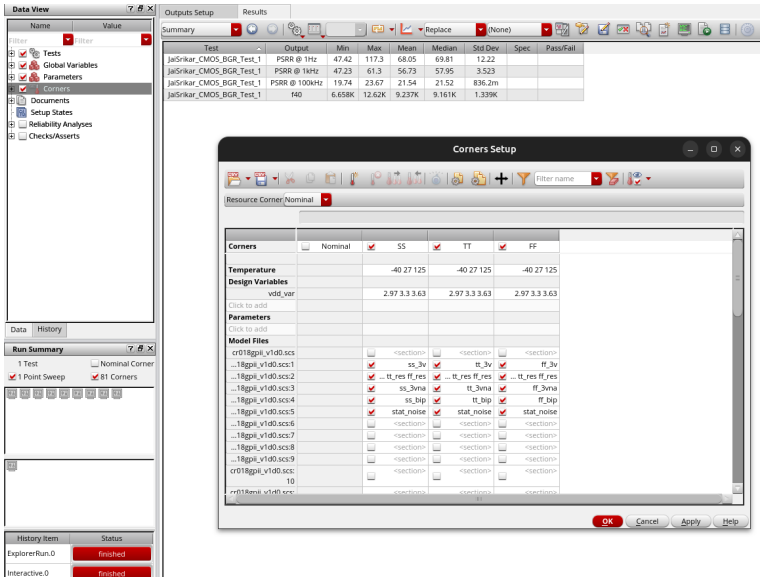


Figure 37: PSRR over 81 corners, summary: 47.4–117.3 dB at 1 Hz, 47.2–61.3 dB at 1 kHz, 19.7–23.7 dB at 100 kHz, 40 dB frequency 6.66–12.62 kHz, with the corner set-up. History `PSRR_PVT`.

7.6 Start-up (Deliverables 15, 17, 22)

History `SU_PVT-only`, 81 corners, 1 pF load (Figures 38, 39 and 40). The circuit starts at every corner. Settling 10.83–44.17 µs: slowest at SS, −40 °C (highest native threshold), fastest at FF, 125 °C. I(MSU) peaks at 36.0–115.3 nA and is at most 2.5 nA afterwards; final V_{REF} 1.249–1.263 V; power 2.99–8.977 µW.

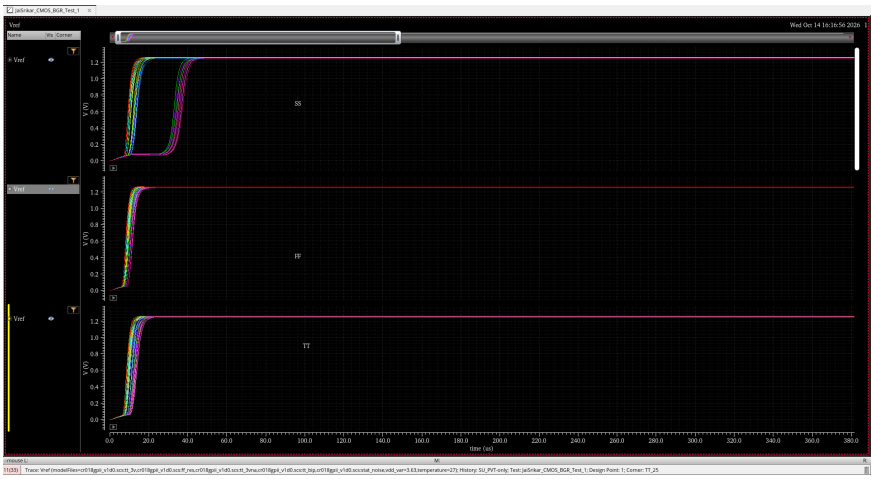
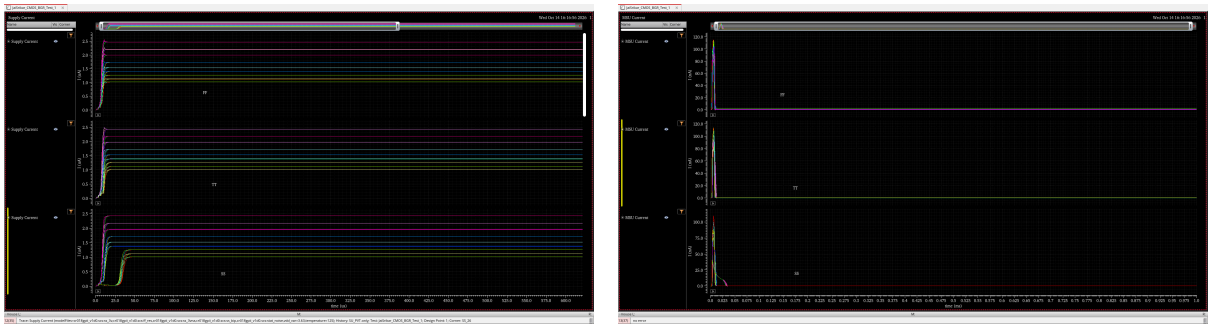


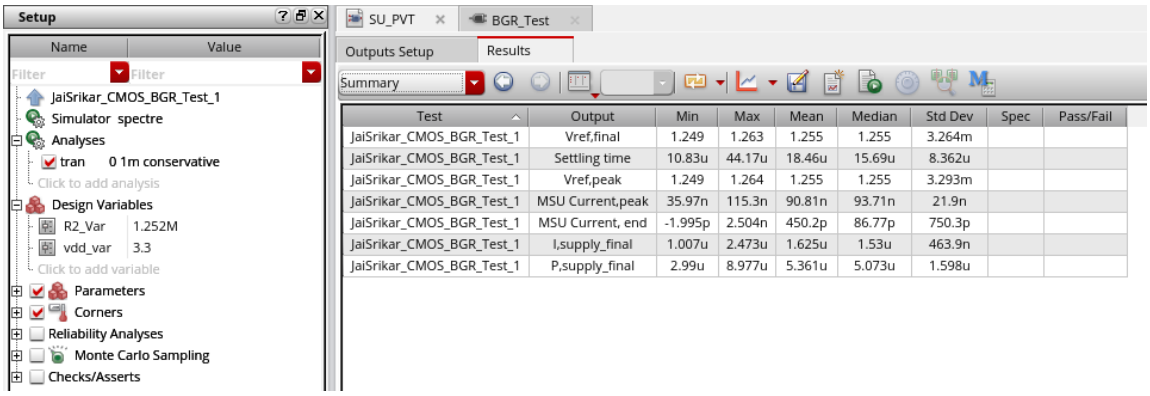
Figure 38: V_{REF} versus time during start-up for all 81 PVT corners, grouped by process corner (27 traces each: 3 resistor corners × 3 supplies × 3 temperatures). The late group in SS is the nine −40 °C corners. History `SU_PVT-only`.



(a) Supply current.

(b) Start-up current I(MSU): peak 36–115 nA, at most 2.5 nA after start-up.

Figure 39: Currents versus time during start-up, 81 corners grouped by process corner. History `SU_PVT-only`.



Test	Output	Min	Max	Mean	Median	Std Dev	Spec	Pass/Fail
JaiSrikar_CMOS_BGR_Test_1	Vref,final	1.249	1.263	1.255	1.255	3.264m		
JaiSrikar_CMOS_BGR_Test_1	Settling time	10.83u	44.17u	18.46u	15.69u	8.362u		
JaiSrikar_CMOS_BGR_Test_1	Vref,peak	1.249	1.264	1.255	1.255	3.293m		
JaiSrikar_CMOS_BGR_Test_1	MSU Current,peak	35.97n	115.3n	90.81n	93.71n	21.9n		
JaiSrikar_CMOS_BGR_Test_1	MSU Current, end	-1.995p	2.504n	450.2p	86.77p	750.3p		
JaiSrikar_CMOS_BGR_Test_1	I,supply_final	1.007u	2.473u	1.625u	1.53u	463.9n		
JaiSrikar_CMOS_BGR_Test_1	P,supply_final	2.99u	8.977u	5.361u	5.073u	1.598u		

Figure 40: Start-up over 81 PVT corners, summary: final V_{REF} 1.249–1.263 V, settling 10.83–44.17 µs, I(MSU) peak 36–115 nA, supply power 2.99–8.977 µW. History `SU_PVT-only`.

7.7 Minimum supply per branch (Deliverable 23)

History `DC_PVT-only`: operating point at VDD = 3.3 V over 27 corners with the expressions of Section 4.4. Left branch (M5) 1.836–2.292 V, middle (M4) 1.665–1.985 V, output (M10) 1.439–1.595 V. The left branch limits at every corner, highest at −40 °C; the worst case leaves 0.68 V of margin at 2.97 V (Figures 41 and 42).

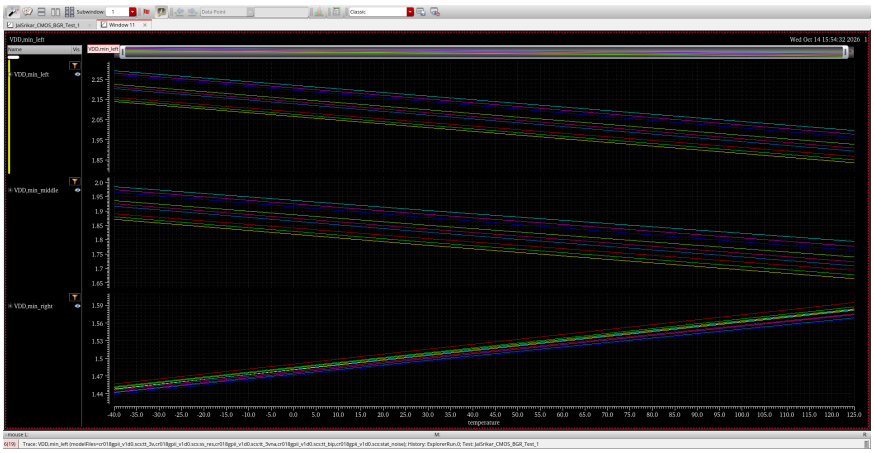


Figure 41: Minimum supply voltage of the left, middle and output branches versus temperature (VDD = 3.3 V, 9 process × resistor combinations each). The left branch (cascode M5) sets the minimum supply at every corner. History `DC_PVT-only`.

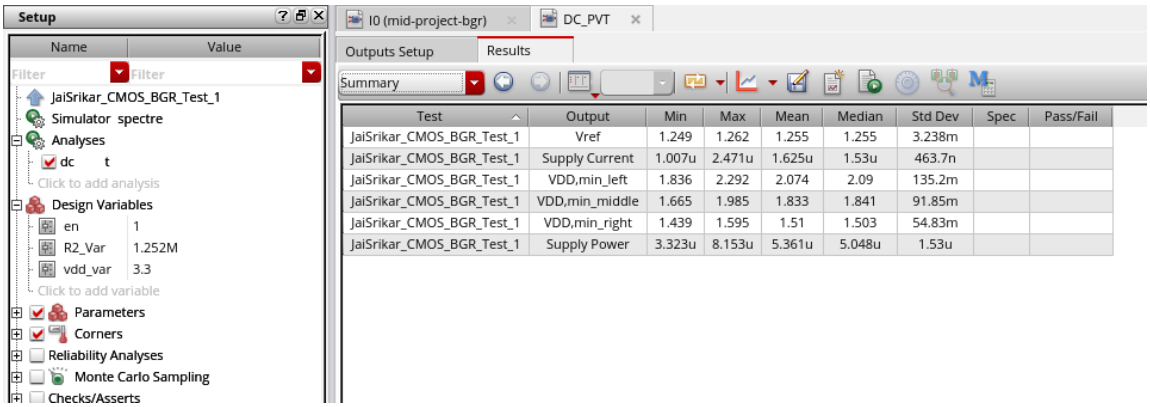


Figure 42: DC operating point over 27 corners, summary: minimum supply left 1.836–2.292 V, middle 1.665–1.985 V, output 1.439–1.595 V. History `DC_PVT-only`.

8 Monte Carlo Results

8.1 Process and mismatch (Deliverables 7, 20)

History `Temp_MC`: 200 samples, tt, 3.3 V, temperature sweep (Figure 43). At 27 °C (Figure 44a) $V_{REF} = 1.257\,\text{V} \pm 3 \times 17.56\,\text{mV}$ (mean $\pm 3\sigma$), all samples within 1.206–1.308 V. The end-point TC has a mean of 1.39 ppm/°C and $\sigma = 11.79$ ppm/°C (Figure 44b); the slopes vary by only 12.3 and 10.9 µV/°C (σ), so the spread is mainly an offset. Maximum power 6.51–8.57 µW.

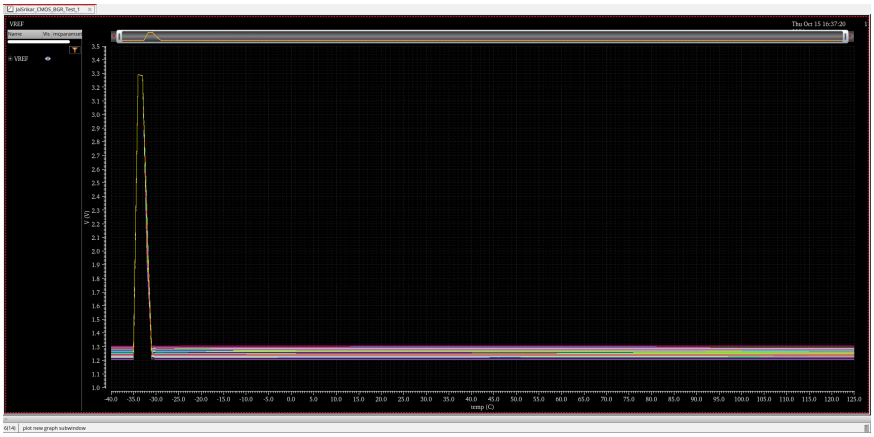
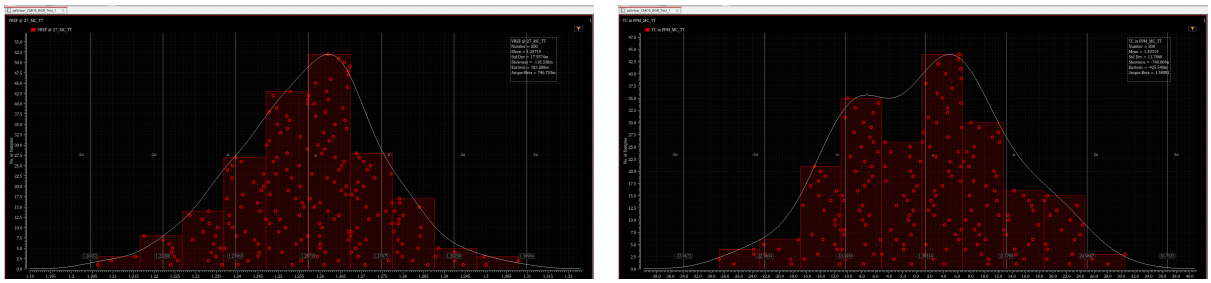


Figure 43: V_{REF} versus temperature, 200 Monte Carlo samples (process and mismatch, tt, 3.3 V). The spike between −36 and −31 °C is the MOS-model artefact. History `Temp_MC`.



(a) V_{REF} at 27 °C: mean 1.257 V, $\sigma = 17.56$ mV. **(b)** End-point TC: mean 1.39 ppm/°C, $\sigma = 11.79$ ppm/°C.

Figure 44: Distributions over 200 Monte Carlo samples (process and mismatch) with the fitted normal curve. History `Temp_MC`.

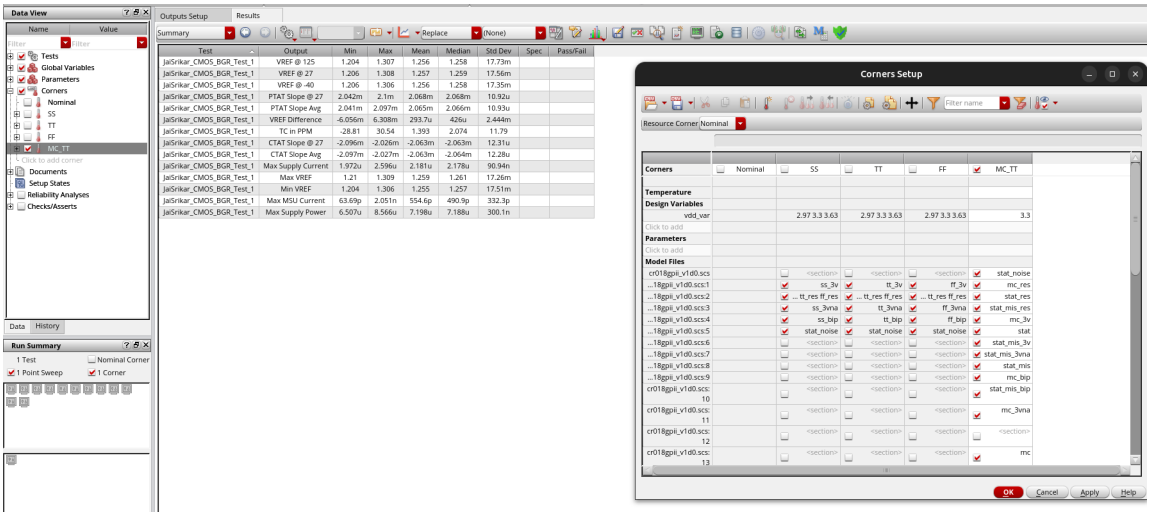


Figure 45: Monte Carlo summary (200 samples) with the MC_TT corner column. History `Temp_MC`.

8.2 Accuracy

Table 14: Accuracy over PVT and Monte Carlo (specification ±8 %).

Quantity	Value	Source
Typical V_{REF} (27 °C)	1.255 V	`Temp_sweep_nom`
PVT minimum / maximum of V_{REF}	1.249 V / 1.263 V	`SU_PVT-only`, `TEMP_PVT`
σ of V_{REF} at 27 °C (200 samples)	17.56 mV ($3\sigma = 52.7$ mV)	`Temp_MC`
Lower limit = min -3σ	1.1963 V → −4.68 %	
Upper limit = max $+3\sigma$	1.3157 V → +4.84 %	

The worst-case variation, −4.68 % / +4.84 %, is inside ±8 %. PVT alone contributes

$-0.48\,\%$ / $+0.64\,\%$; the mismatch term ($\pm 4.2\,\%$) dominates.

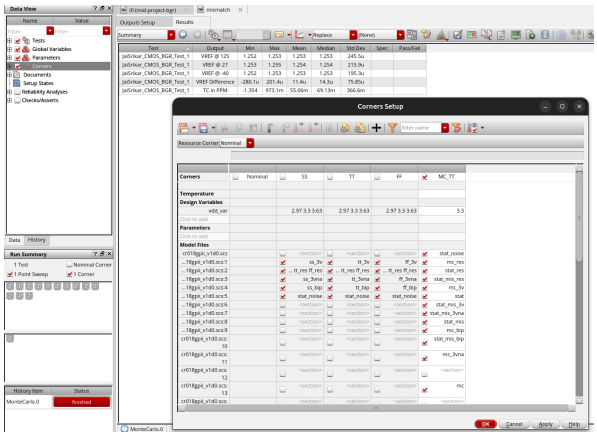
8.3 Mismatch-only contributions (Deliverables 11–14)

Four runs of 200 samples with statistical variation "Mismatch", each with one group selected: Q1–Q3; R1, RA, RB and the three segments of R2; M1–M4; M5–M10 (the PMOS selection also contains MINVP, MPU1 and MPU2, which are off in operation).

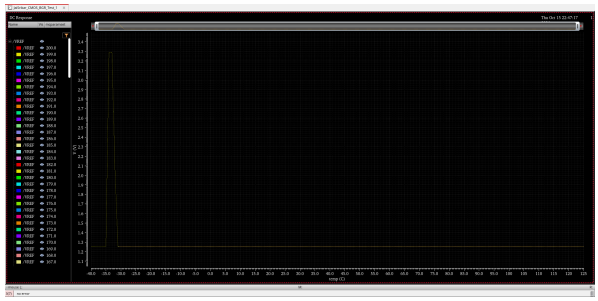
Table 15: Mismatch-only Monte Carlo, V_{REF} at 27 °C.

Group (history)	$\sigma(V_{REF})$	Range	Share of variance
BJTs Q1–Q3 (`BJT-only-mismatch`)	0.216 mV	1.253–1.255 V	0.02 %
Resistors (`RES-only-mismatch`)	0.302 mV	1.253–1.255 V	0.03 %
NMOS M1–M4 (`NMOS-only-mismatch`)	11.11 mV	1.226–1.283 V	40 %
PMOS M5–M10 (`PMOS-only-mismatch`)	12.67 mV	1.221–1.287 V	52 %

The MOS mirrors dominate; the RSS of the four groups (16.86 mV) is close to the full 17.56 mV, the PMOS value agrees with the hand estimate (12.4 mV), and the NMOS estimate (14.5 mV) is conservative.

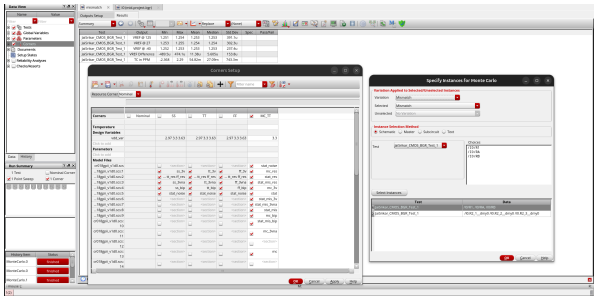


(a) Summary.

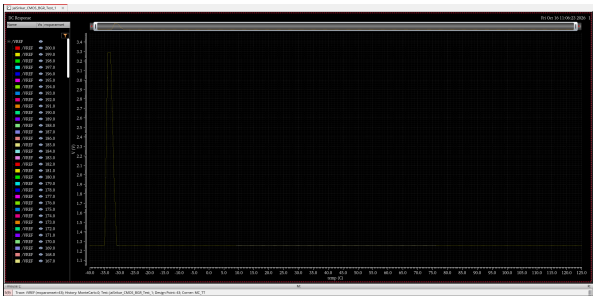


(b) V_{REF} versus temperature.

Figure 46: Mismatch-only Monte Carlo, BJTs Q1–Q3 (200 samples, tt, 3.3 V): $\sigma(V_{REF}$ at 27 °C) = 0.216 mV. History `BJT-only-mismatch`.

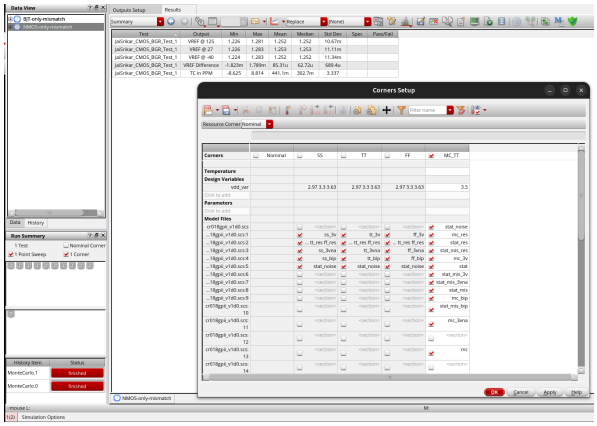


(a) Summary with the instance list.

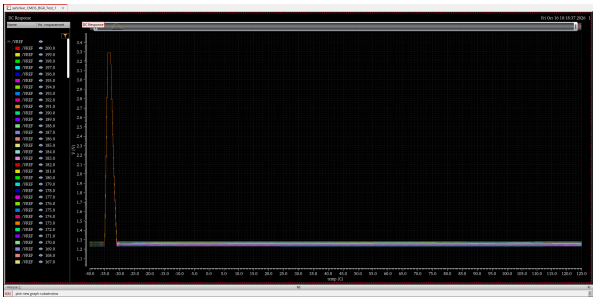


(b) V_{REF} versus temperature.

Figure 47: Mismatch-only Monte Carlo, resistors R1, R2 (three segments), RA, RB: $\sigma = 0.302\,\text{mV}$. History `RES-only-mismatch`.

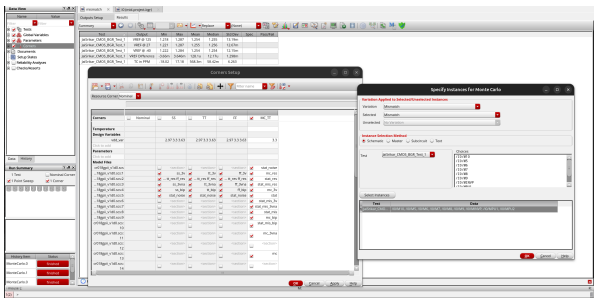


(a) Summary.

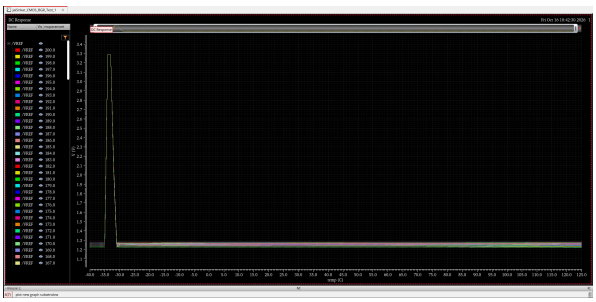


(b) V_{REF} versus temperature.

Figure 48: Mismatch-only Monte Carlo, NMOS mirror and cascodes M1–M4: $\sigma = 11.11\,\text{mV}$. History `NMOS-only-mismatch`.



(a) Summary with the instance list.

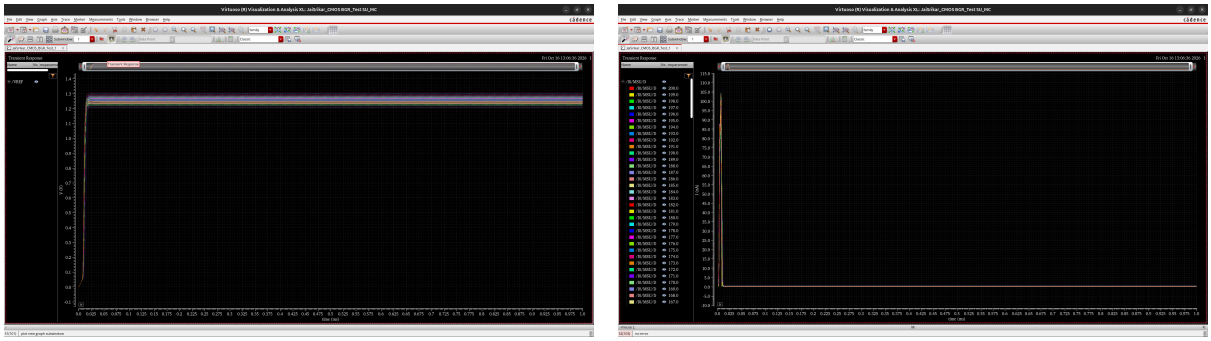


(b) V_{REF} versus temperature.

Figure 49: Mismatch-only Monte Carlo, PMOS mirror and cascodes M5–M10 (the selection also contains the inverter and pull-up PMOS, which are off in operation): $\sigma = 12.67\,\text{mV}$. History `PMOS-only-mismatch`.

8.4 Start-up Monte Carlo (Deliverables 16, 18)

History `SU_MC_ONLY`: 200 samples, tt, 27 °C, 3.3 V, 1 pF load. Every sample starts (Figure 50); settling 14.7–20.5 µs (mean 15.72 µs), I(MSU) peak 75.2–104.8 nA, then 7–597 pA; final V_{REF} 1.206–1.308 V; power 4.54–6.11 µW (Figure 51).



(a) V_{REF}: every sample starts and settles between 1.21 and 1.31 V.

(b) Start-up current I(MSU): peak 75–105 nA within ≈ 20 µs, then pA leakage.

Figure 50: Start-up, 200 Monte Carlo samples (process and mismatch, tt, 27 °C, 3.3 V, 1 pF load). History `SU_MC_ONLY`.

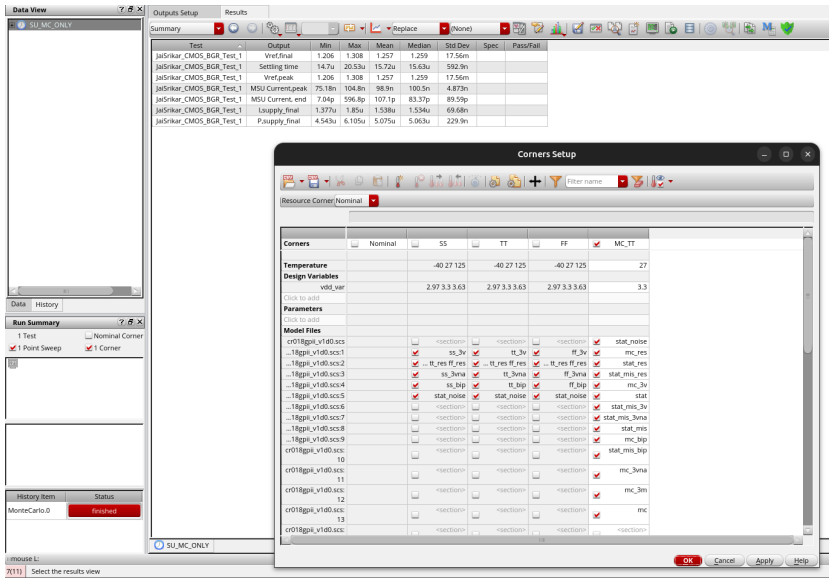


Figure 51: Start-up Monte Carlo (200 samples), summary: settling 14.7–20.5 µs, I(MSU) peak 75–105 nA, I(MSU) at 1 ms 7–597 pA, final V_{REF} 1.206–1.308 V, power 4.54–6.11 µW. History `SU_MC_ONLY`.

9 Conclusion

The cascoded beta-multiplier bandgap reference delivers 1.255 V with an end-point temperature coefficient of $-0.068\,\text{ppm}/^\circ\text{C}$, consumes 5.05 µW (42 pA in power-down) and starts in 16.11 µs at the typical point. Over all 81 PVT corners the power is at most 8.977 µW ($< 10\,\mu\text{W}$), the PSRR at 1 Hz above 47.4 dB and the circuit starts within 44.17 µs. The worst-case variation is $-4.68\,\%$ / $+4.84\,\%$, inside the $\pm 8\,\%$ specification.

References

[1] B. Razavi, *Design of Analog CMOS Integrated Circuits*. New York: McGraw-Hill Education, 2nd ed., 2017.

A Deliverables Map

Table 16: Deliverables 1–23: where they are shown.

#	Deliverable	Section	Figures	Key value
1	V_{EB} vs. temperature across P, V	7.1	30a, 18	CTAT slope −2.102...−2.013 mV/°C
2	$(\Delta V_{EB}/R_1)R_2$ vs. temperature across P, V	7.1	30b, 18	PTAT slope 2.049–2.120 mV/°C
3	Slope expressions across PVT	5.3, 7.1	33, 19	Table 12
4	Supply current vs. temperature across P, V	7.2	31, 18	max 2.473 µA (8.977 µW)
5	Power-sequencing transients	6.7	26, 27	starts on every EN pulse
6	V_{REF} vs. temperature across P, V	7.3	32, 33	1.249–1.263 V
7	Monte Carlo accuracy / ppm	8.1, 8.2	44, 43, 45	σ 17.56 mV; −4.68 % / +4.84 %
8	Line-sensitivity plot, 0–4 V	7.4	34, 20	–
9	Line-sensitivity expression and value	5.3, 7.4	35, 21	0.375–2.505 mV/V
10	PSRR across PVT	7.5	36, 37	$\geq$ 47.4 dB at 1 Hz
11	Mismatch-only MC, BJTs	8.3	46	σ 0.216 mV
12	Mismatch-only MC, resistors	8.3	47	σ 0.302 mV
13	Mismatch-only MC, NMOS	8.3	48	σ 11.11 mV
14	Mismatch-only MC, PMOS	8.3	49	σ 12.67 mV
15	V_{REF} vs. time across PVT	7.6	38, 24	1.249–1.263 V final
16	V_{REF} vs. time, Monte Carlo	8.4	50, 51	all 200 samples start
17	Start-up current vs. time across PVT	7.6	39	peak 36.0–115.3 nA
18	Start-up current vs. time, Monte Carlo	8.4	50b, 51	peak 75.2–104.8 nA
19	V_{REF} expression across PVT	2.1, 7.3	33, 40	Eq. (2); Table 13
20	V_{REF} expression, Monte Carlo	8.1	44	1.257 V ± 3 × 17.56 mV
21	Settling without start-up	6.8	28, 29	stays at 0.33 V, never settles
22	Settling with start-up across PVT	7.6, 8.4	40, 25	10.83–44.17 µs
23	Minimum supply per branch across P, T	4.4, 7.7	41, 42	left branch 1.836–2.292 V